Explainable Machine Learning

Generalized additive models

ETSIINF
Universidad Politécnica de Madrid
b.mihaljevic@upm.es



GAM

A generalized additive model (GAM) has the form

$$g(\mathrm{E}(Y \mid x_1, x_2, \ldots, x_n)) = \alpha + f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n),$$

where

- $g(\cdot)$ is a link function, such as the identity function in linear regression with Gaussian errors;
- the f_j are typically non-parametric functions.

The f_j terms are single-feature models also called shape functions.

Interpretation

- The f_j are non-parametric: we lose coefficients β from linear models.
- They can interpreted, however, by visually inspecting their graph.

GA^{2}M

Pairwise interactions can be added to standard GA^{2}M, leading to a model called *Generalized Additive Models plus Interactions*, GA^{2}M,

$$g(\mathrm{E}(Y \mid X_1, X_2, \ldots, X_n)) = \alpha + \sum_{j=1}^{n} f_j(x_j) + \sum_{i \neq j} f_{ij}(x_i, x_j).$$

Interpretation

Pairwise interactions can be visualized as a heat map.

GAM / GA^{2}M

- Since the shape functions can be arbitrarily complex, GA^{2}M overcome some restrictions of linear models.
- On many problems, their prediction performance is comparable to that of complex models such as random forest and boosting.
- Interpretable, modular, and editable.

Fitting shape functions

- Smoothing splines
- Decision trees with gradient boosting and bagging (Explainable Boosting Machine)
- Neural nets

Table of Contents

1. Case study: Pneumonia risk prediction
2. Learning with splines
3. Learning with trees
4. Summary

Pneumonia risk

Predict the probability of death (POD) for patients with pneumonia so that high-risk patients could be admitted to the hospital while low-risk patients be treated as outpatients.

GAM and GA^{2}M accurate

Model	Pneumonia
Logistic Regression	0.8432
GAM	0.8542
GA^{2}M	0.8576
Random Forests	0.8460
LogitBoost	0.8493

- GAM and GA^{2}M best, albeit small differences in AUC.
- Note: logistic regression actually non-linear due to feature discretization.

Pneumonia risk

Data

- 14,199 pneumonia patients
- 1542 (10.86%) patients died from pneumonia
- Train set: 9847; test set: 4352 (70/30 train/test split)
- 46 features

- Baseline probability of death: 10.86%.
- Patients with a predicted probability of death above baseline considered high-risk.

Pneumonia risk

Patient-history findings			
chronic lung disease	-	age	C
re-admission to hospital	-	gender	-
admitted through ER	-	diabetes mellitus	-
admitted from nursing home	-	asthma	-
congestive heart failure	-	cancer	-
ischemic heart disease	-	number of diseases	C
cerebrovascular disease	-	history of seizures	-
chronic liver disease	-	renal failure	-
history of chest pain	-		
Physical examination findings			
diastolic blood pressure	C	wheezing	-
gastrointestinal bleeding	-	stridor	-
respiration rate	C	heart murmur	-
altered mental status	-	temperature	C
heart rate	C		
Laboratory findings			
liver function tests	-	BUN level	C
glucose level	C	creatinine level	C
potassium level	C	albumin level	C
hematocrit	C	WBC count	C
percentage bands	C	pH	C
pO2	C	pCO2	C
sodium level	C		
Chest X-ray findings			
positive chest x-ray	-	lung infiltrate	-
pleural effusion	-	pneumothorax	-
cavitation/empyema	-	chest mass	-
lobe or lung collapse	-		

Ranging from history features such as age and gender, to simple measurements such as heart rate, to lab tests such as Blood Urea Nitrogen (BUN), to features read from a chest x-ray such as lung collapse.

Pneumonia risk

Binary classification GAM (aka additive logistic regression)

The probability of death is given by

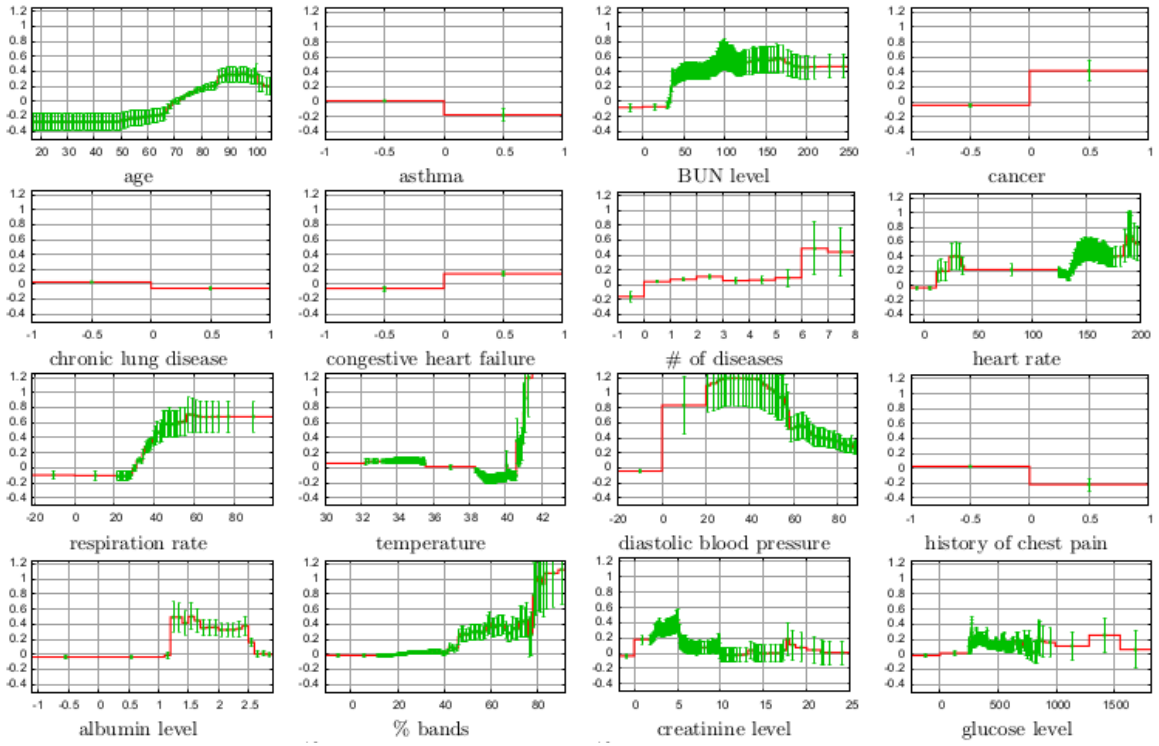
$$g(\mathrm{E}(Y \mid x_1, x_2, \ldots, x_n)) = r(x_1, \ldots, x_n) = \alpha + f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n),$$

where

- α is the intercept;
- Y is Bernoulli and $\mathrm{E}(Y \mid x_1, \ldots, x_n) = P(Y = 1 \mid x_1, \ldots, x_n) = \mathrm{POD}$;
- $r(x_1, \ldots, x_n)$ is the total risk score, log-odds of POD, $\frac{P(Y=1|x_1,x_2,\ldots,x_n)}{P(Y=0|x_1,x_2,\ldots,x_n)}$.
- $g(p) = \log \frac{p}{1-p}$ is the logit function.

Pneumonia risk

Shape functions



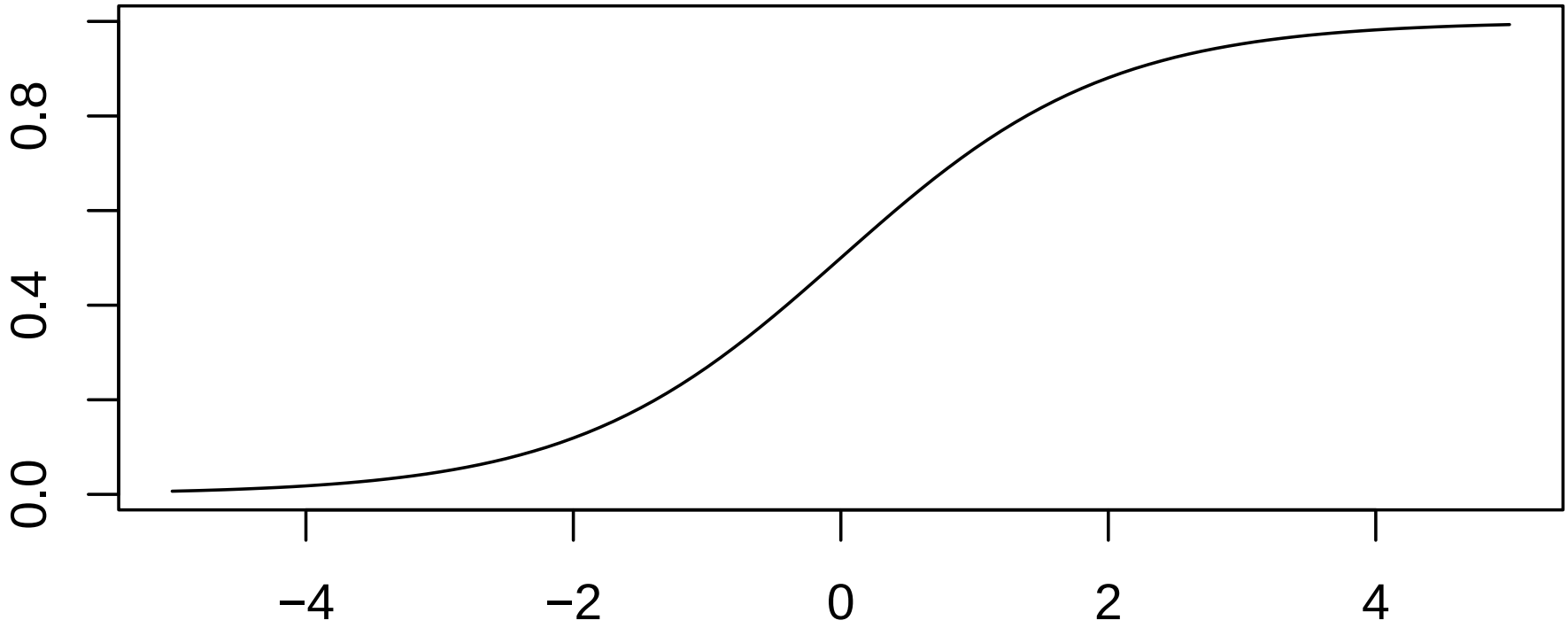
Intercept

The logit of the base rate (prior probability, $\mathrm{E}(Y) = P(Y = 1)$): $g(0.1086) = -2.105$.

Pneumonia risk

For a given patient

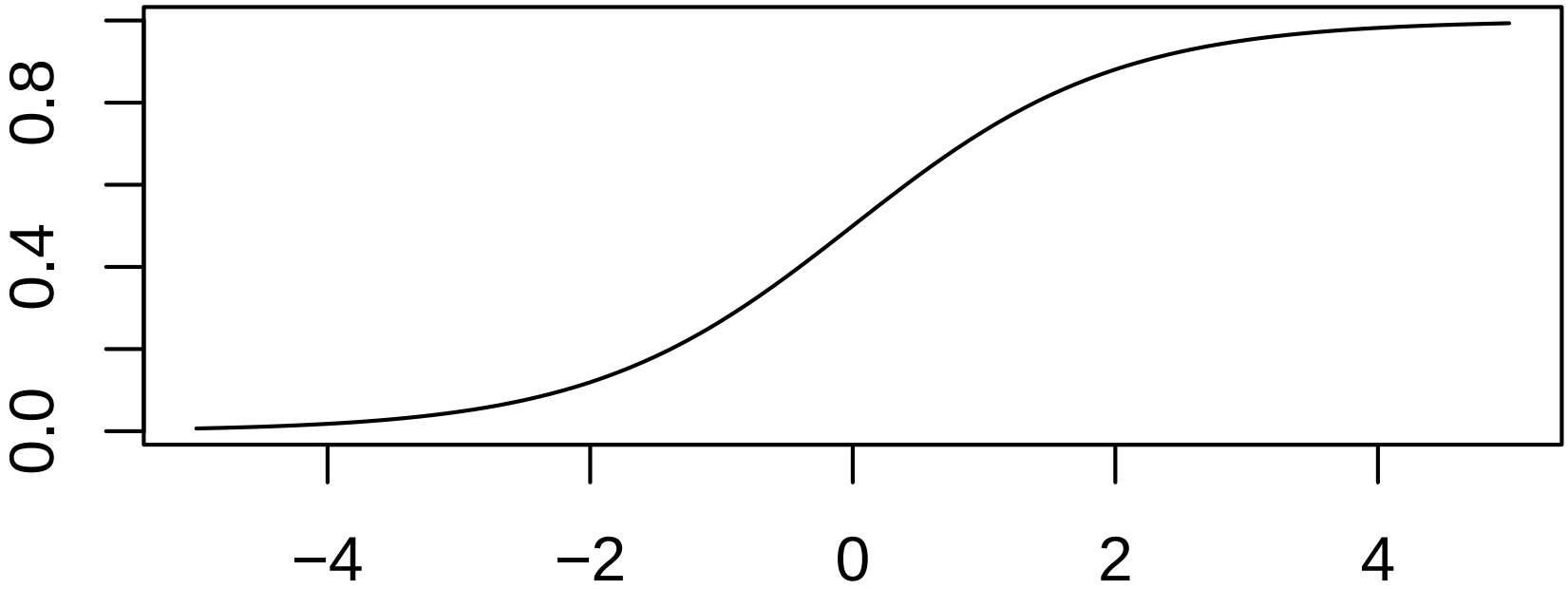
- The total risk, $r(x_1, \ldots, x_n) = \alpha + \sum_{j=1}^{n} f_j(X_j) = -2.105 + \sum_{j=1}^{n} f_j(X_j)$ converted to probability of death with $g^{-1}(r(x)) = 1/\left(1 + e^{-r(x)}\right)$.
- For example, a patient with total risk $r(x_1, \ldots, x_n) = 0$ has a high probability of death: $g^{-1}(0) = 0.5$.
- The probability of death is monotonic, non-linear in risk.



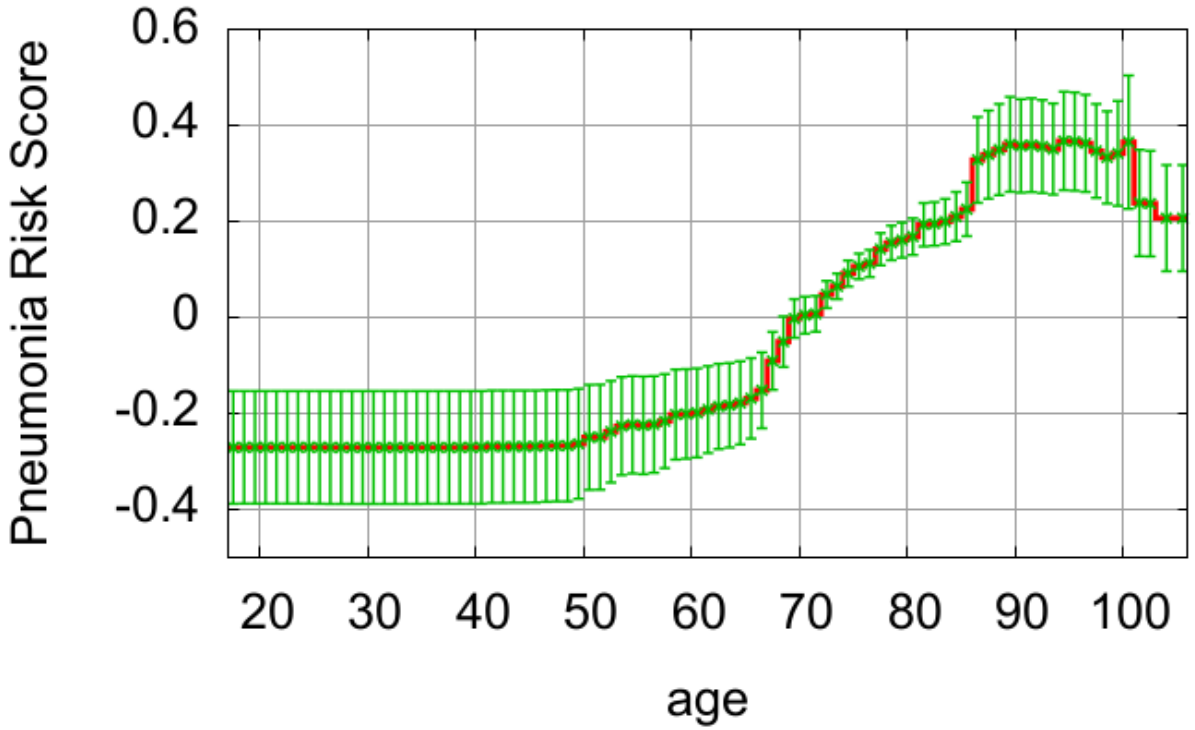
Pneumonia risk

For a given patient

- Total risk above the baseline (-2.105) means above average probability of death.
- That is, if the terms' contributions to $r(x_1, \ldots, x_n)$ are positive, the probability of death is above average.
- If all the terms evaluate to 0, the patient has average probability of death.

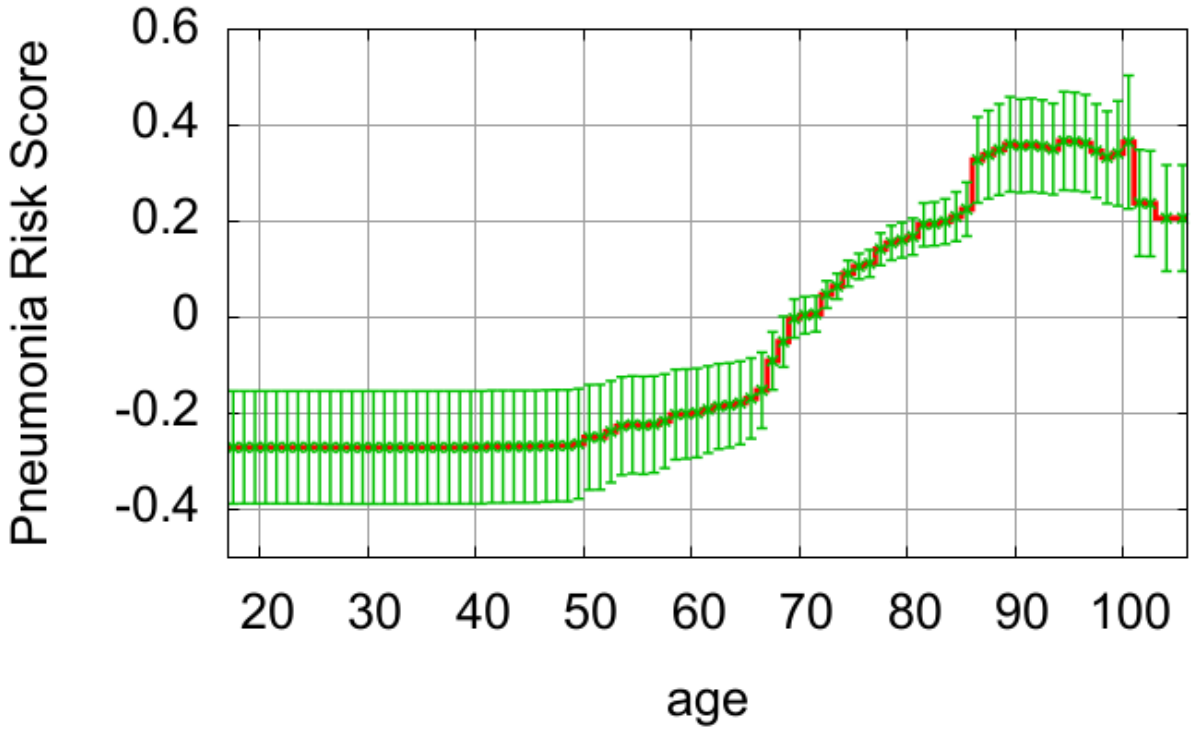


Pneumonia risk



- Each graph is centered vertically so that the mean prediction over the training set is 0.
- This is done by subtracting the mean score for the feature.

Pneumonia risk



- Terms with scores above (below) zero increase (decrease) total risk.
- Being 80 years old adds to risk whereas having 20 subtracts from it.
- Thus, while each term f_j is a risk score (log odds), it is interpreted with respect to the baseline.

Centering terms

1. Fit the terms f_j.
2. Center each f_j by subtracting the average score.
3. Add an intercept α; this makes the average prediction $=$ baseline.

Baseline

- Binary classification: logit of the prior (e.g., -2.3 if the prior is 10%).
- Regression: Average Y value.

Notes

- We are not centering the feature X_j but rather the output $f_j(X_j)$.
- All averages are over the training set.
- If we did not add the intercept, the average prediction would be 50% for positive class in binary classification (and 0 in regression).

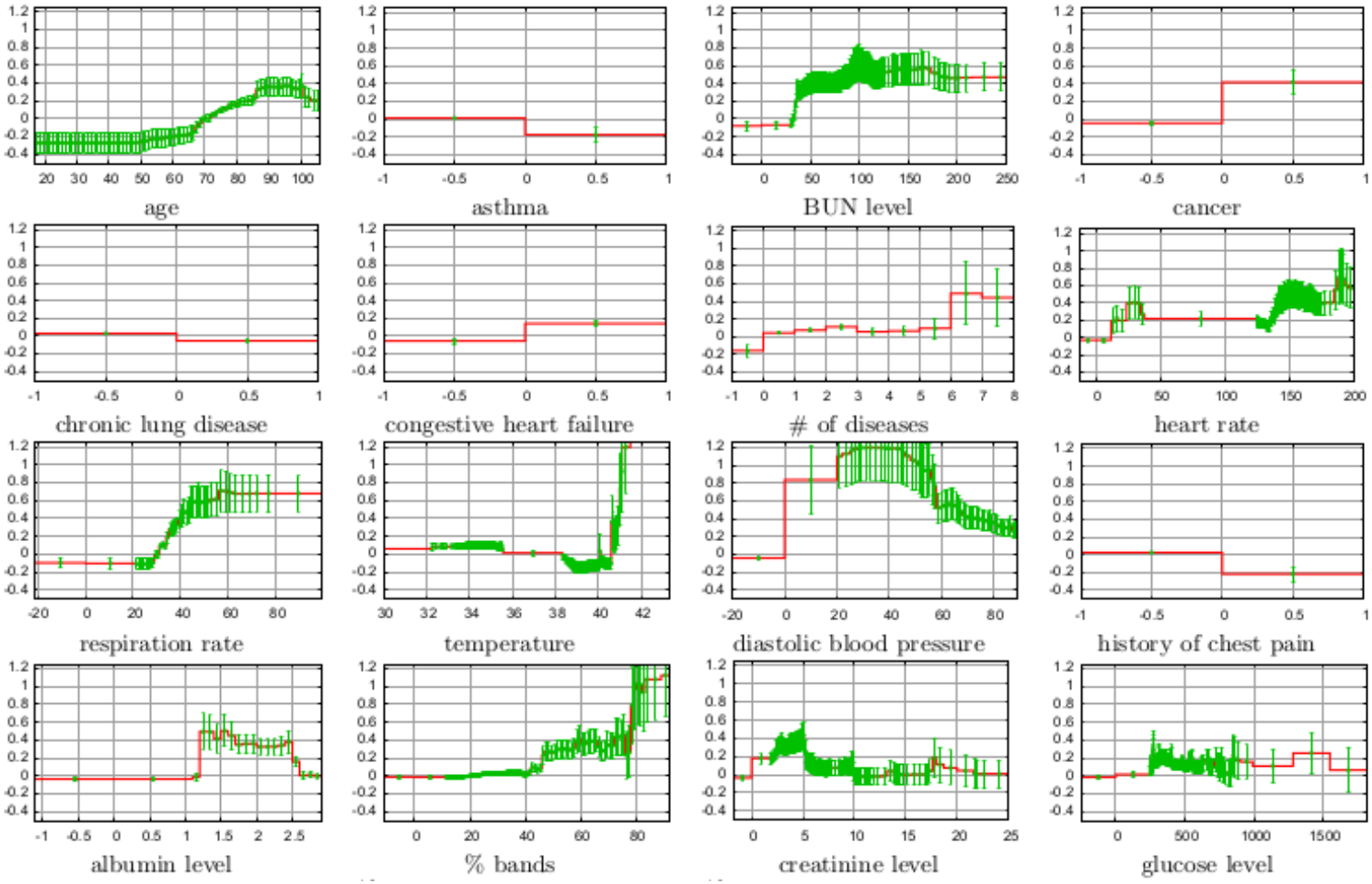
Modularity

Centering terms makes the model identifiable and modular.

Removing / adding terms

- We might be interested in, for example, using expert knowledge to modify the model.
- Each term can be removed from the model without affecting the intercept: the intercept estimator ignores whether a term is included or not.
- If all terms were removed, only the intercept would remain, and the prediction would be the baseline.
- Likewise, adding new terms to the model does not alter the intercept.

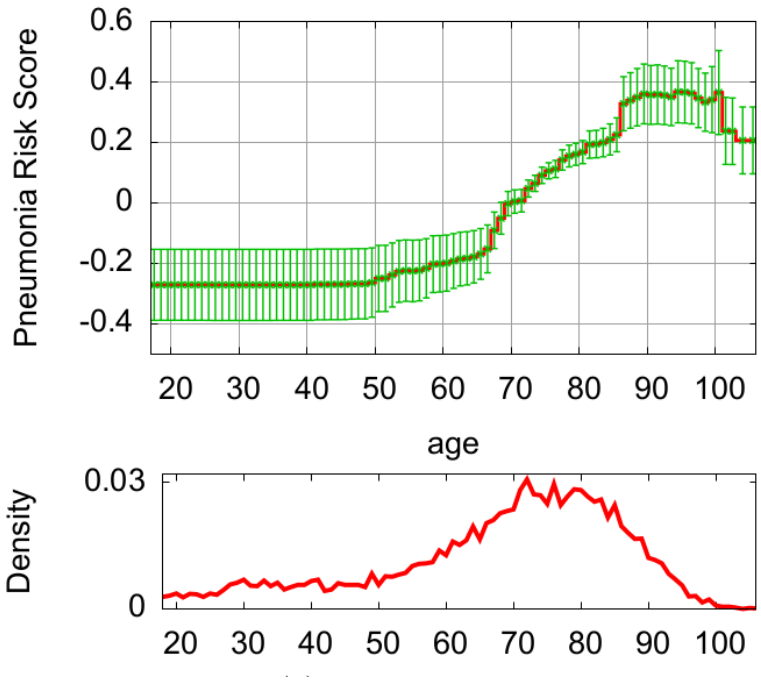
Pneumonia risk



- Each graph is centered vertically: mean prediction over training set is 0.
- Same Y-axis scale on all graphs for ease of comparison.

Pneumonia risk

Age



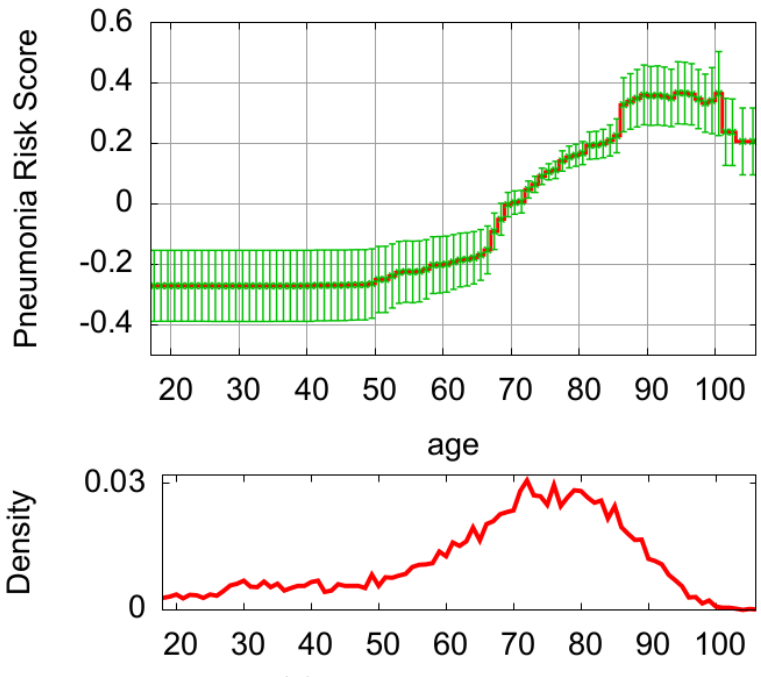
- X-axis: Ranges from 18 to 106 years old (only adults).
- Y-axis: Predicted risk score as a function of age.
- Risk score: from ≈ -0.25 for age < 50 to ≈ 0.35 for age > 85.
- Age has risk scores as high as 0.35.
- Consistent with previous knowledge: Pneumonia is sometimes called "The Old Man's Best Friend".

Note: the actual contribution to POD due to, e.g., $f_{\text{Age}}(x) = 0.35$, depends on the baseline and the values of other terms; recall that logistic function is non-linear.

Pneumonia risk

Age: Inspecting the function

Most patients are age 60-90 – where the error bars are tightest.

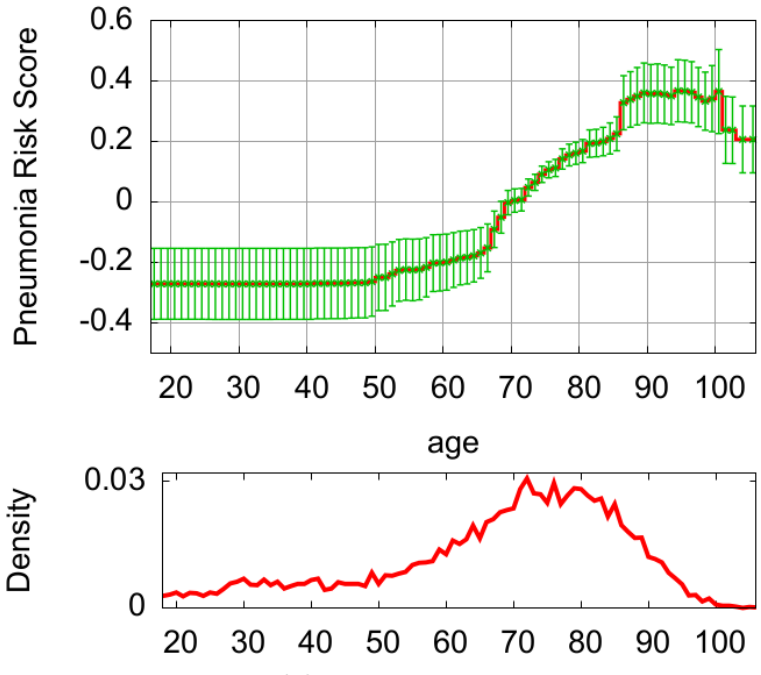


Risk by age:

- 18-50: Constant, negative– -0.27.
- 50-66: Rises slowly, still negative.
- 66-90: Rises quickly; 0.0 at 70, +0.20 at 82, +0.35 at 86.
- 90-100: Levels off at very high risk.
- Above 100: Apparent drop in risk.

Pneumonia risk

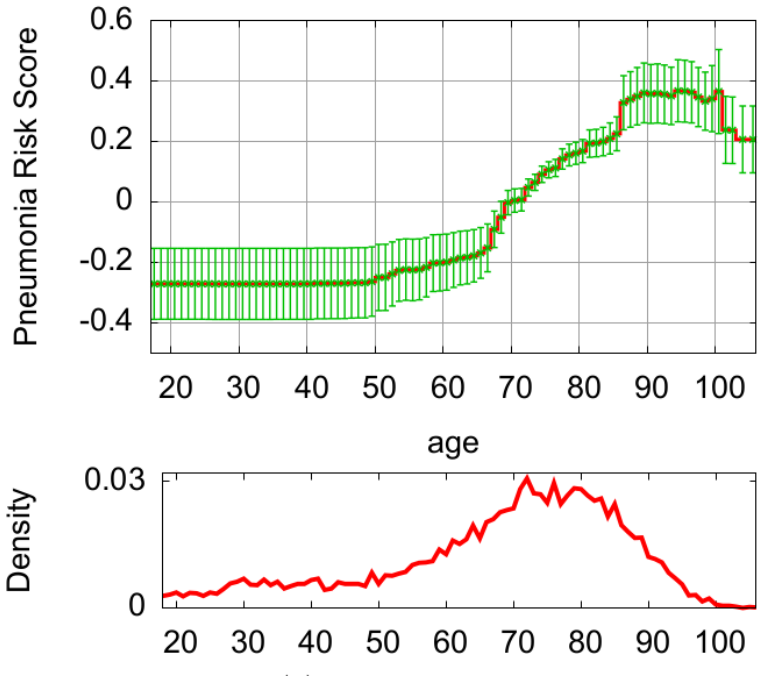
Age: Inspecting the function – Interpreting



- Risk is only positive after age 70 and no evidence that it further increases above age 85.
- The increase in risk of going from 70 to 86 is larger than the decrease in risk of going from 70 down to 50 or less.
- Jumps in risk at age 67, and again at age 86.

Pneumonia risk

Age: Hypothetical explanations



Negative risk for age 18-50:

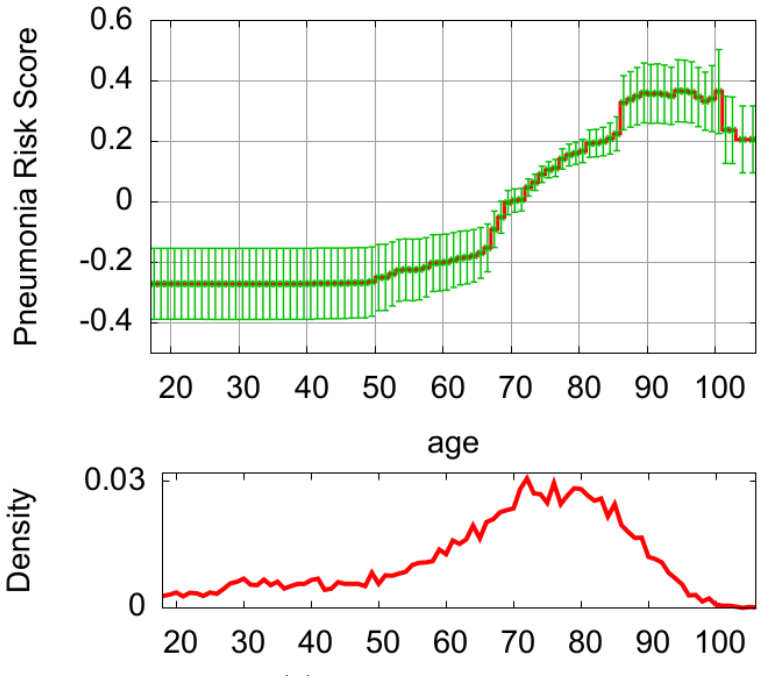
- Being young significantly reduces the risk of dying from pneumonia.

Small jump in risk at age 67:

- Tight error bars – jump may be real.
- Many patients in the 90's would have retired at around age 65.
- Retirement may yield differences in activity levels, health insurance, and willingness to get access to healthcare early enough to improve outcomes.

Pneumonia risk

Age: Hypothetical explanations

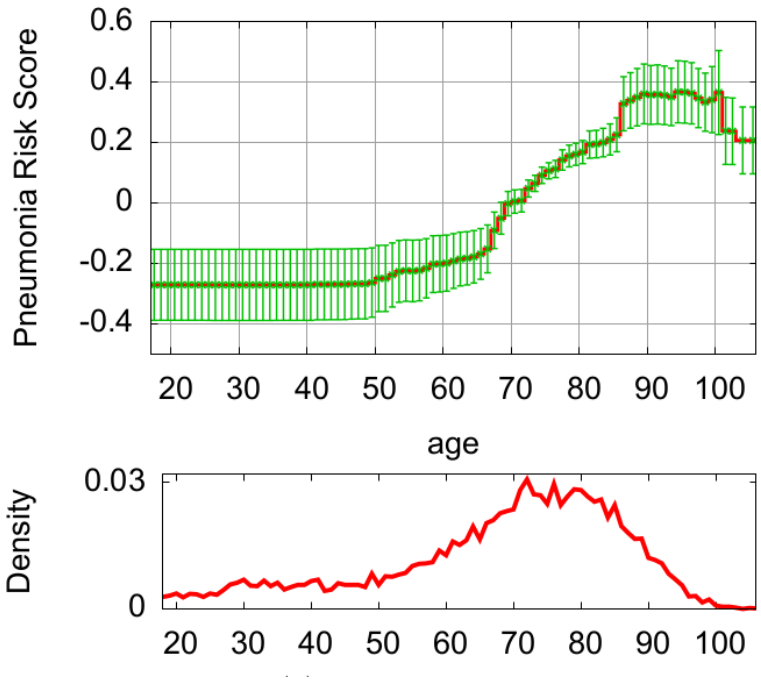


Large jump in risk around age 86:

- Hard to explain.
- Perhaps practitioners treat patients older than 85 differently, and this ultimately increases their POD.
- Perhaps artifact of the model – the error bars large around age 86.

Pneumonia risk

Age: Hypothetical explanations



Apparent drop in risk above age 100.

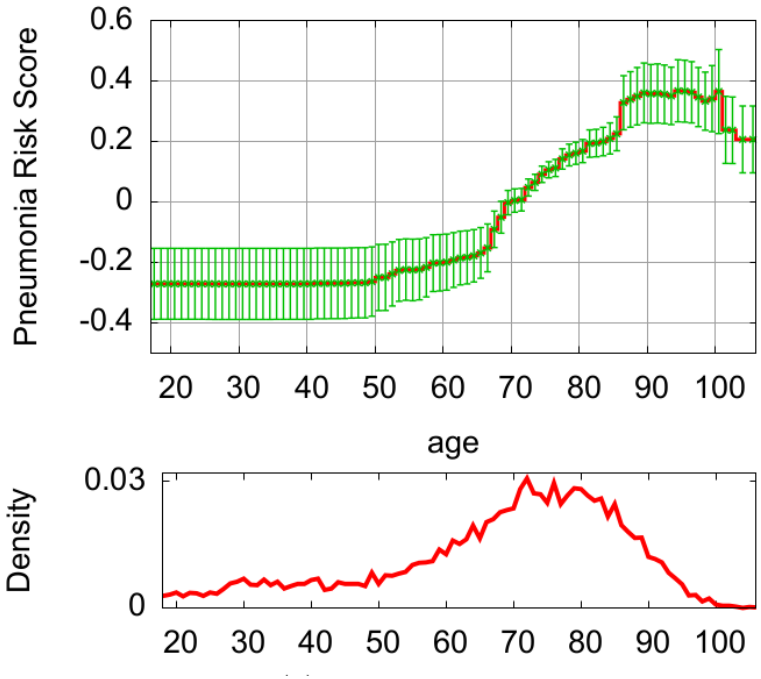
- Perhaps artifact of the model – few patients above age 95, error bars are large and consistent with risk being constant.

No evidence that risk increases after 85.

- Perhaps medical treatments equally effective for patients older than 85.
- Perhaps other medical conditions more likely to be responsible for death at this age than pneumonia.
- Perhaps model artifact – failed to learn it.

Pneumonia risk

Interpretation

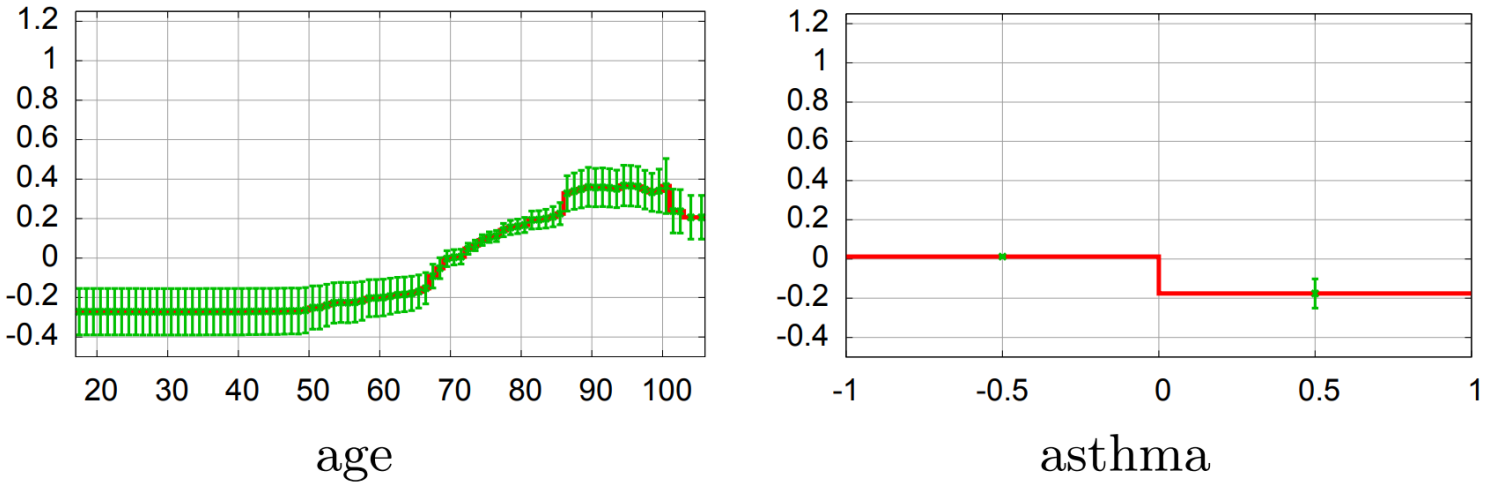


The errorbar is ± 1 standard deviation of the variation in the risk score measured by 100 rounds of bagging.

For each term, we can also show:

- data density for different values of the feature,
- descriptive statistics,
- how they are measured,
- what the normal ranges are,
- what abnormal values indicate,

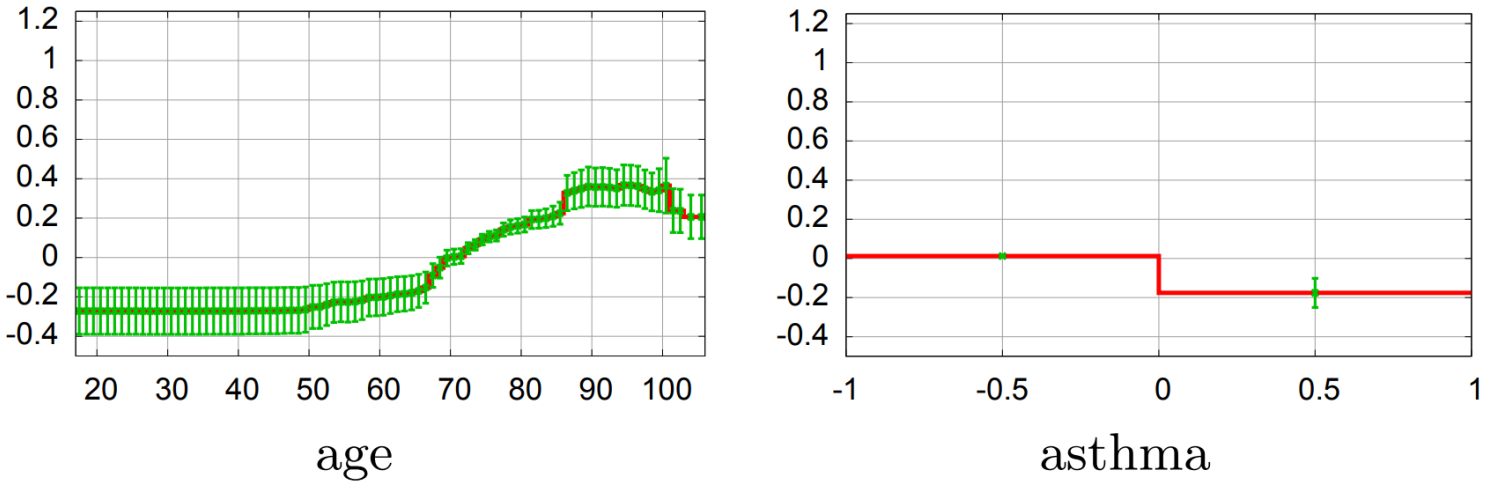
Pneumonia risk



Asthma

- X-axis: Absence (0) or presence (1) of asthma history.
- Y-axis: Predicted risk score as a function of asthma.
- Counterintuitive: pneumonia patients with a history of asthma have a risk score slightly below baseline.

Pneumonia risk



Asthma

- A true pattern: asthmatic patients who presented with pneumonia usually were admitted directly to the ICU. Aggressive care received at ICU was so effective that it lowered their risk of dying from pneumonia compared to the general population.
- However, ICU not in the model: not a variable that would be available at predict time.
- History of asthma gets confounded with outcome, *in the opposite direction*.

Predict pneumonia

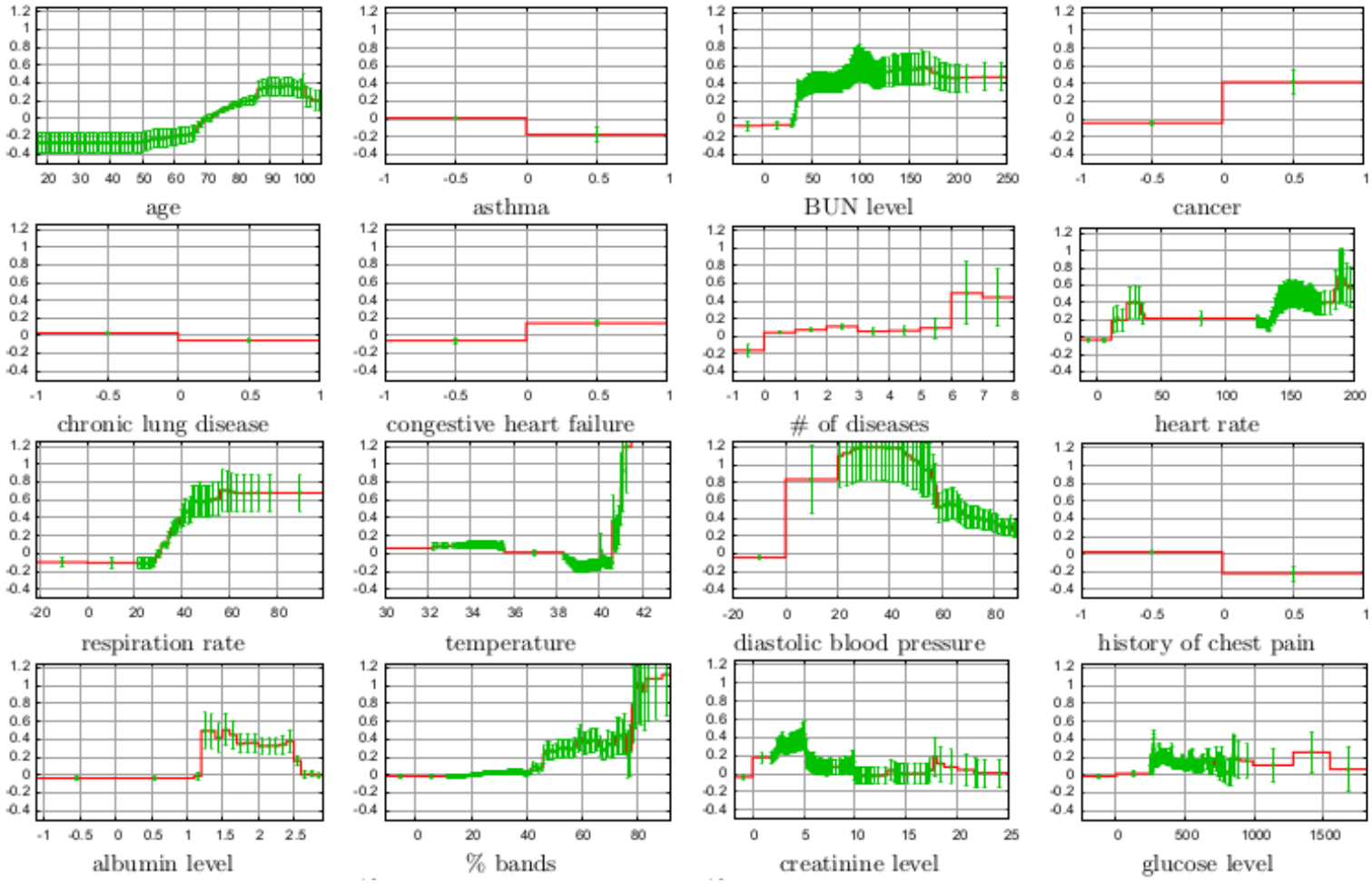
Debugging

- This relation between asthma and probability of death prevented clinical trials with the accurate neural net model in the 90's study.
- Unlike in neural nets, however, the relation is easy to both **recognize** and **fix** in a GAM/GA^{2}M model.
- This is because the model is interpretable and modular.

Correcting

- Eliminate the term (making $f_j = 0$ for all x)
- Using human expertise to redraw the graph, ensuring that the risk score for asthma=1 is positive, not negative.

Term importance



Term importance

Global

- If there are many terms, unfeasible for experts to examine all carefully.
- We lack per-feature coefficients of linear models.
- Need to rank terms by importance in order to prioritize inspection.

Sort with measures such as:

- Mean absolute risk score;
- Drop in AUC when the term is removed;
- Performance of the term when used in isolation;
- Maximum contribution (positive or negative) that the term can make for any patient;
- ...

Term importance

Local

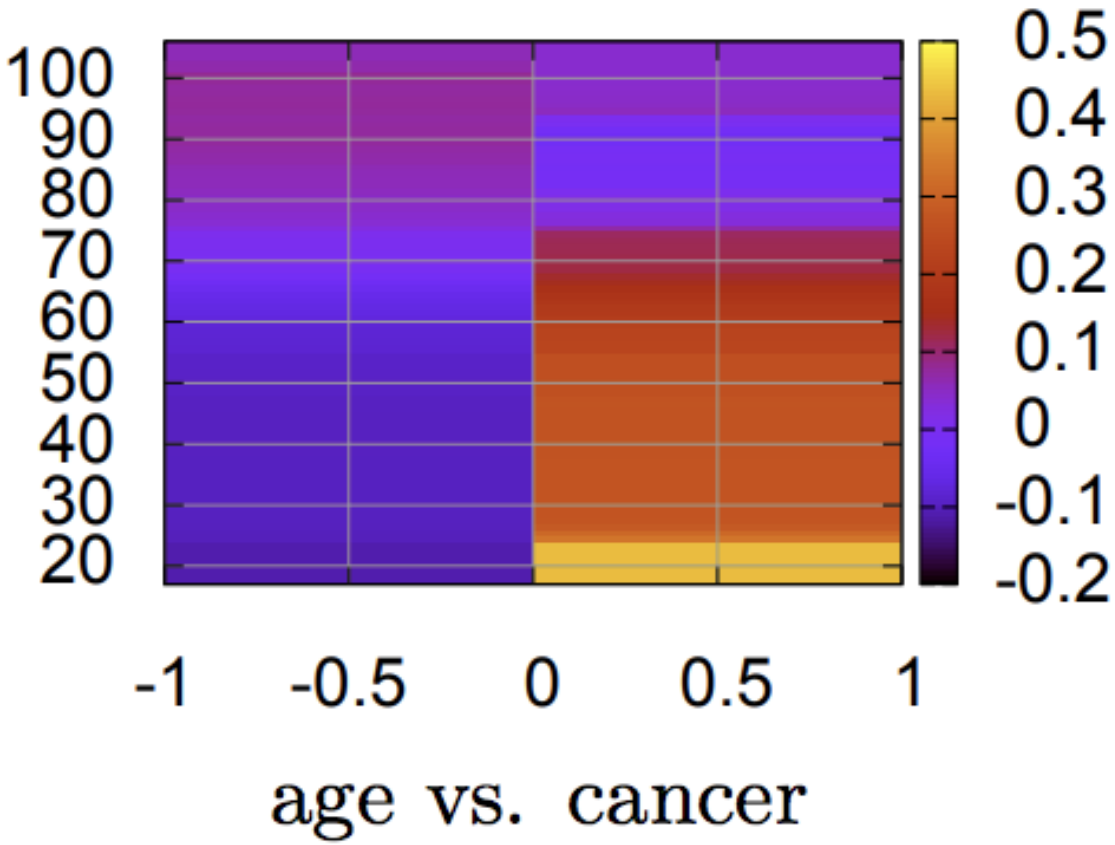
- Easy to sort terms by importance when predicting for a single instance.
- Each term yields a single risk score for an instance.
- Sort terms by how much they increase or decrease risk for the patient.
- Thus identify key characteristics that best explain the prediction.
- Even with large models, few terms may be of importance for most instances; e.g., in a hospital readmission prediction model with above 4000 terms often less than 100 were relevant.

GA^{2}M

Pairwise interactions can be added to standard GA^{2}M, leading to a model called GA^{2}M,

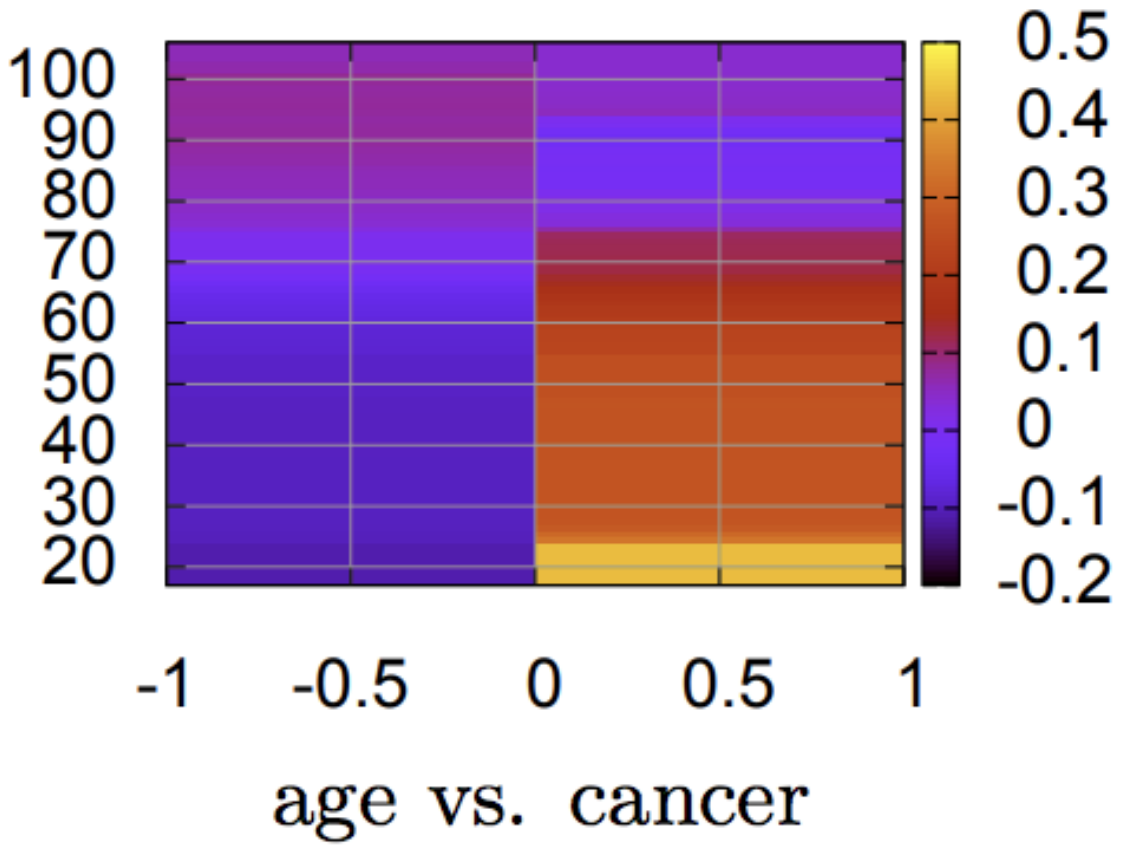
$$g(\mathrm{E}(Y \mid X_1, X_2, \ldots, X_n)) = \alpha + \sum_{j=1}^{n} f_j(x_j) + \sum_{i \neq j} f_{ij}(x_i, x_j).$$

GA^{2}M



- Risk highest for youngest patients: probably cancers acquired in childhood and not cured by age 18.
- Interaction: risk declines with age for patients with cancer, but rises with age for patients without cancer.
- Effect likely due to difference between childhood and adult cancers.

GA^{2}M



- Build GAM first, then choose k pairs of interactions in the residuals.
- The interaction term $f_{jk}(X_j, X_k)$ does not replace the univariate terms $f_j(X_j)$ and $f_k(X_k)$.
- Center $f_{jk}(X_j, X_k)$ so that the average risk score is zero.

GA^{2}M

Model	Pneumonia
Logistic Regression	0.8432
GAM	0.8542
GA^{2}M	0.8576
Random Forests	0.8460
LogitBoost	0.8493

- GA^{2}M had 56 terms, adding 10 pairwise interactions to the GAM model, yet without much improvement.
- Note: AUC, instead of error rate, as the classes are imbalanced (10% positive class).

Feature shaping vs. expert discretization

Model	Pneumonia
Logistic Regression	0.8432
GAM	0.8542
GA^2M	0.8576
Random Forests	0.8460
LogitBoost	0.8493

- An to alternative shape functions: experts discretizing features.
- Experts carefully discretized each continuous attribute into clinically meaningful ranges; e.g., the age intervals were 18-39, 40-54, 55-63, 64-68, 69-72, 73-75, 76-79, 80-83, 84-88, and 89+.
- However, GAMs outperformed logistic regression with discretized features indicating that feature shaping is valuable.
- Also, GA^2M with discretized features was slighly worse than plain GA^2M.

Summary

- The interpretable model uncovered surprising patterns in the data that previously had prevented complex learned models from being fielded in this domain.
- Because it is intelligible and modular, it allowed for recognizing and removing these patterns.

Table of Contents

1 Case study: Pneumonia risk prediction

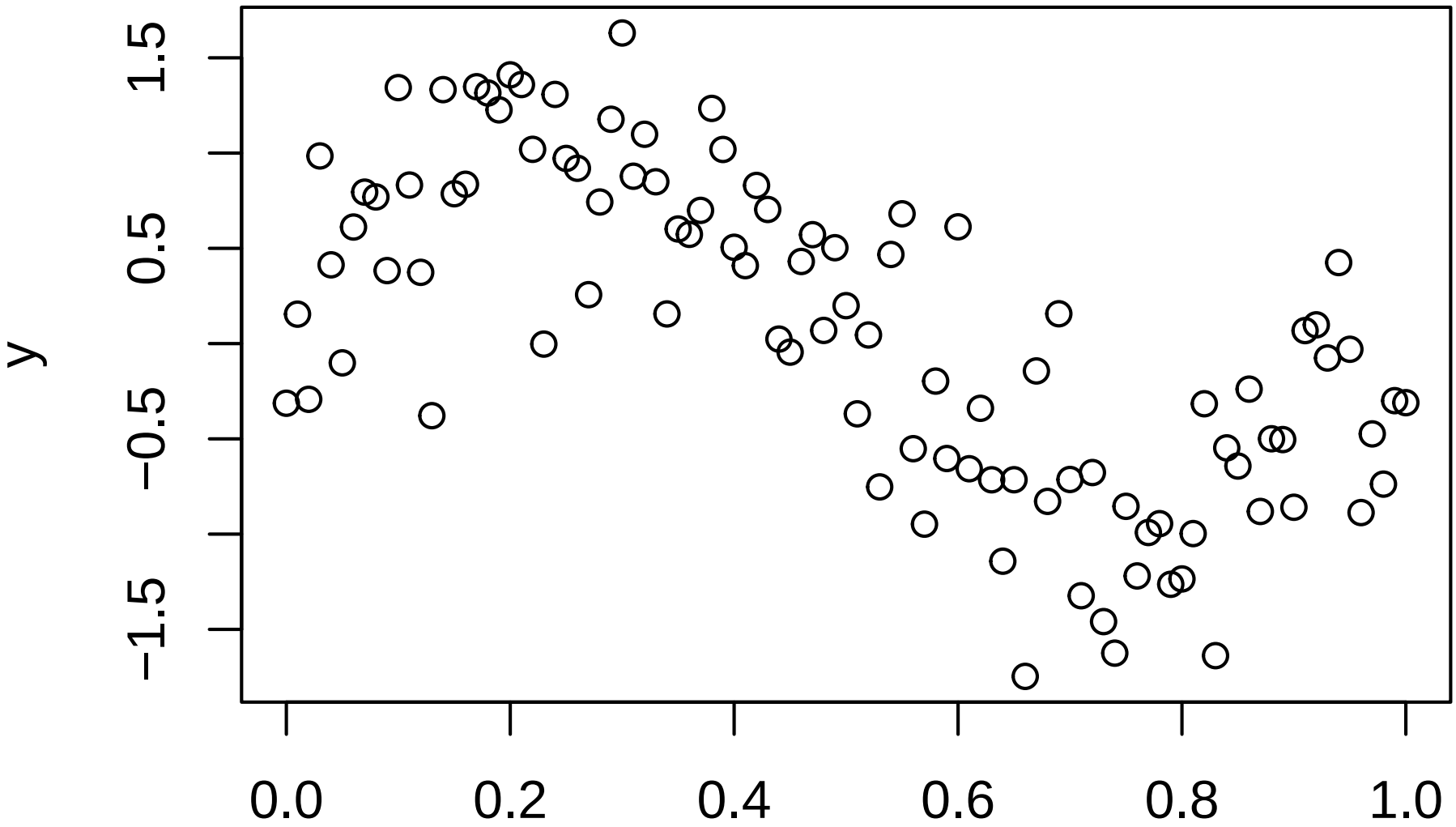
2 Learning with splines

3 Learning with trees

4 Summary

Splines

Consider modelling Y as a non-linear function of a one-dimensional X.

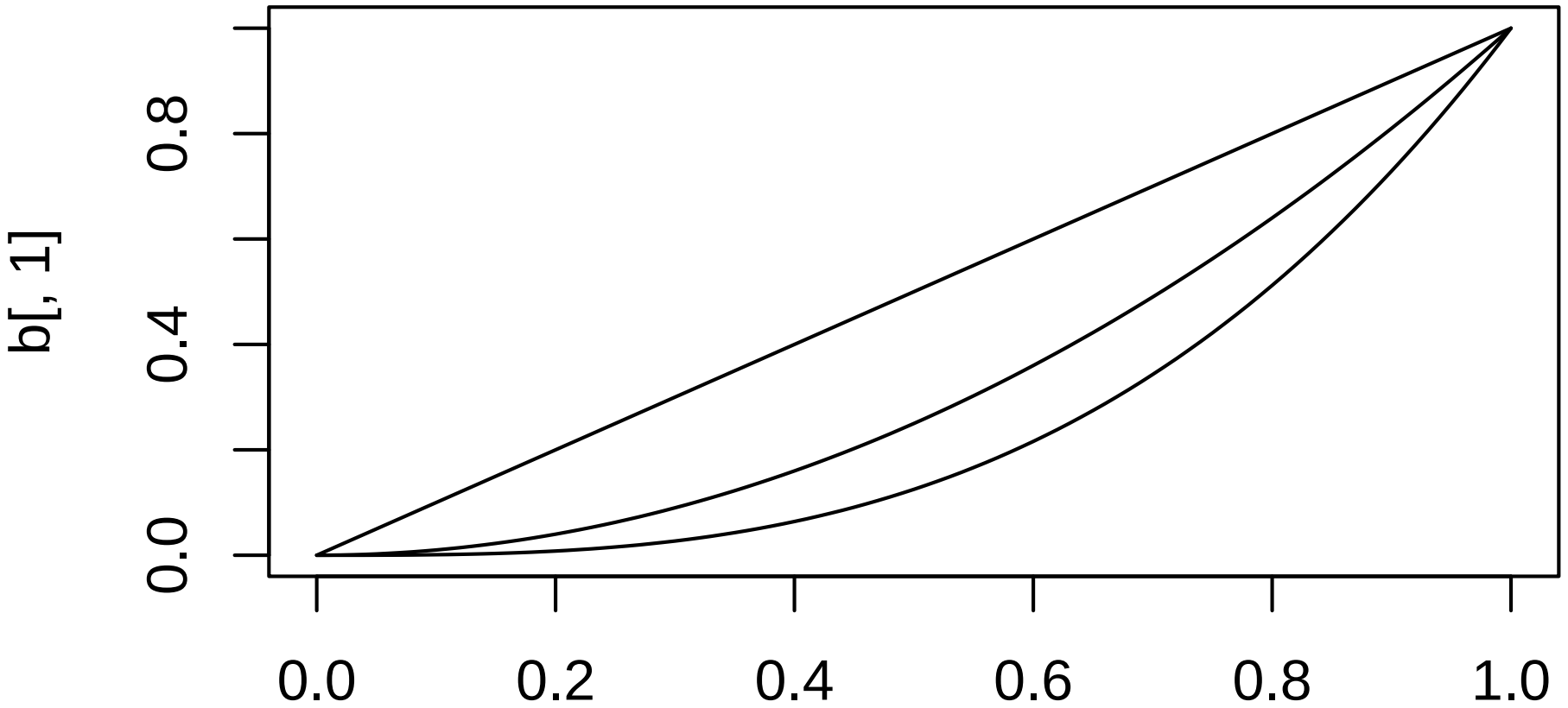


Splines

- One option is to model Y as a degree m polynomial in X.
- We can do this by expanding X into $X, X^2, X^3, \ldots X^m$ and fit a linear model in this new basis.

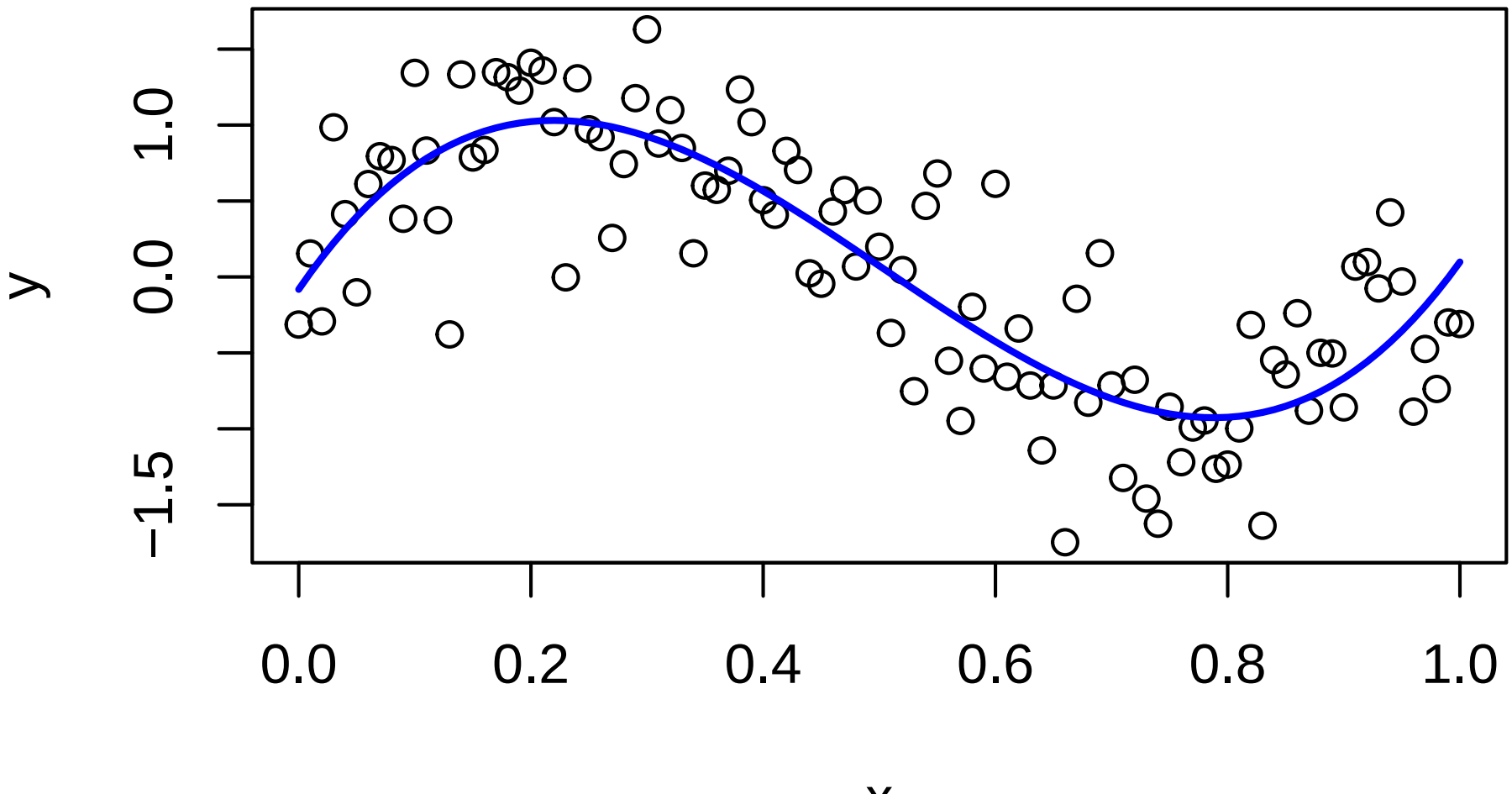
Splines

```
b <-  poly(x,3, raw = TRUE)
plot(x, b[, 1], type = 'l', ylim = c(0, 1))
lines(x, b[, 2])
lines(x, b[, 3])
```



Splines

```
plot(x, y)
p <- predict(lm(y ~ b))
lines(x, p, lwd = 2, col = 'blue')
```

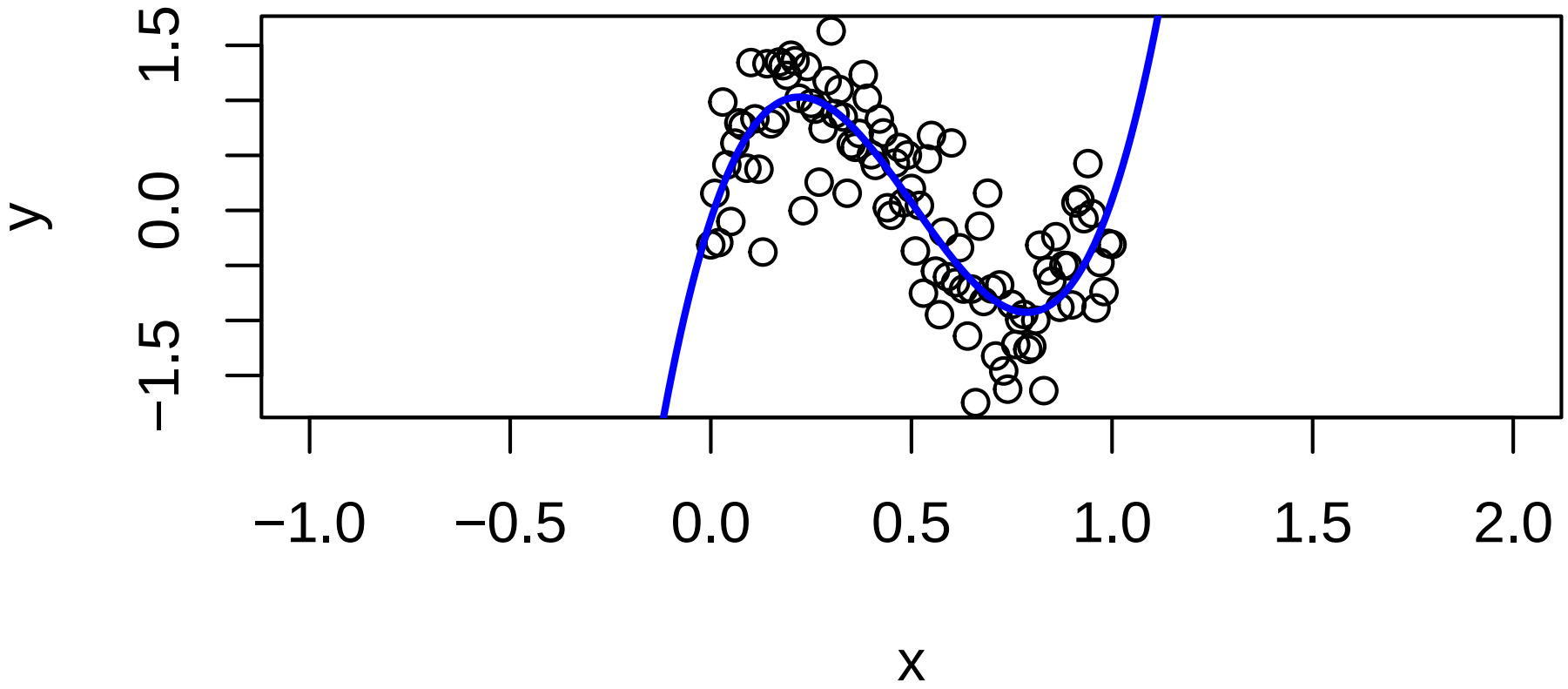


Splines

```
##
## Call:
## lm(formula = y ~ b)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.24487 -0.29504 -0.01103  0.31573  1.03860
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.08207    0.17638  -0.465    0.643
## b1           11.14312    1.53515   7.259 9.73e-11 ***
## b2          -32.37156    3.57736  -9.049 1.52e-14 ***
## b3           21.40917    2.35092   9.107 1.14e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 '
```

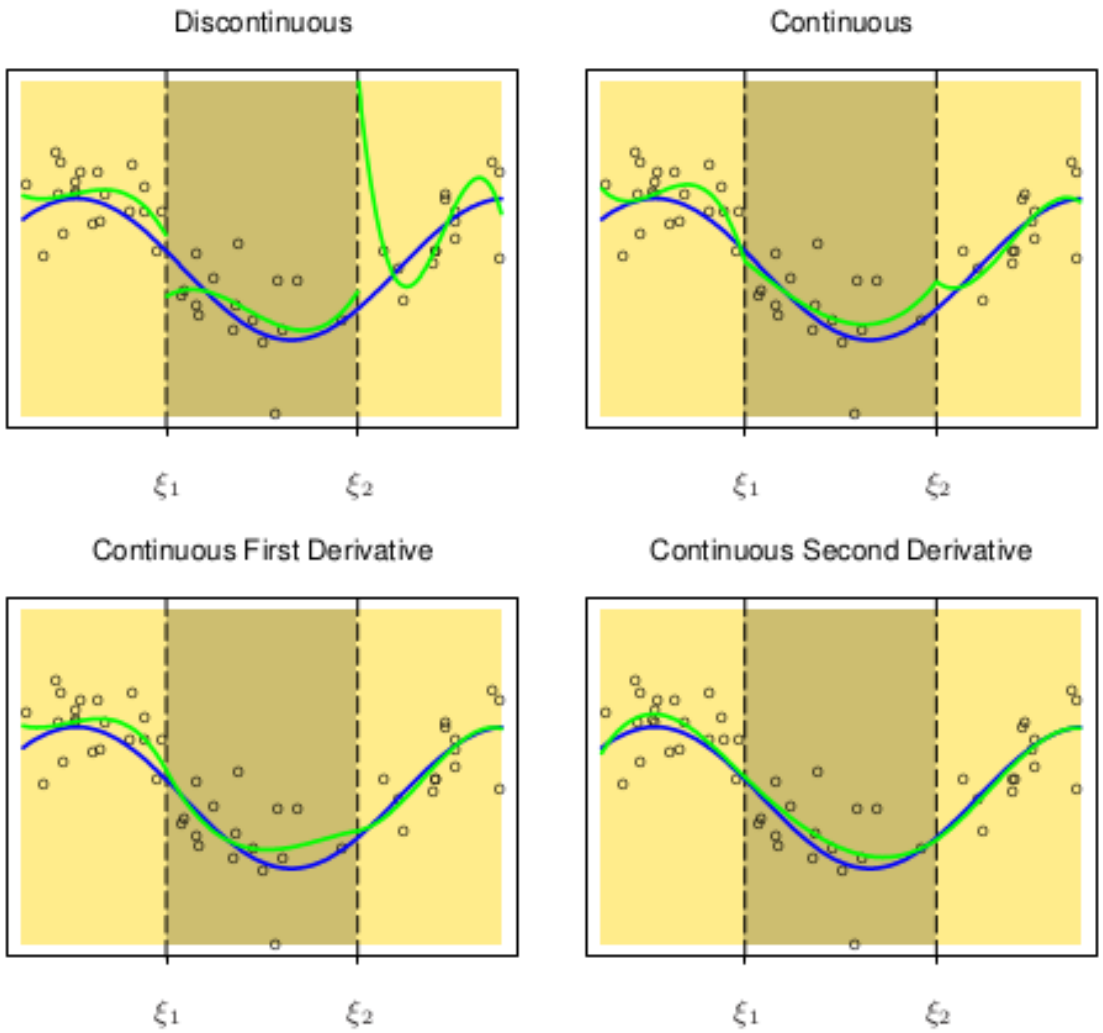
Splines

- Still linear in coefficients: `lm()` fit it with ordinary least squares.
- However, the global nature of polynomials is problematic: adjusting coefficients to achieve a specific form in one region can cause dramatic changes in the function in remote regions.



Piecewise polynomials

An alternative is to fit piecewise polynomials.



Piecewise polynomials

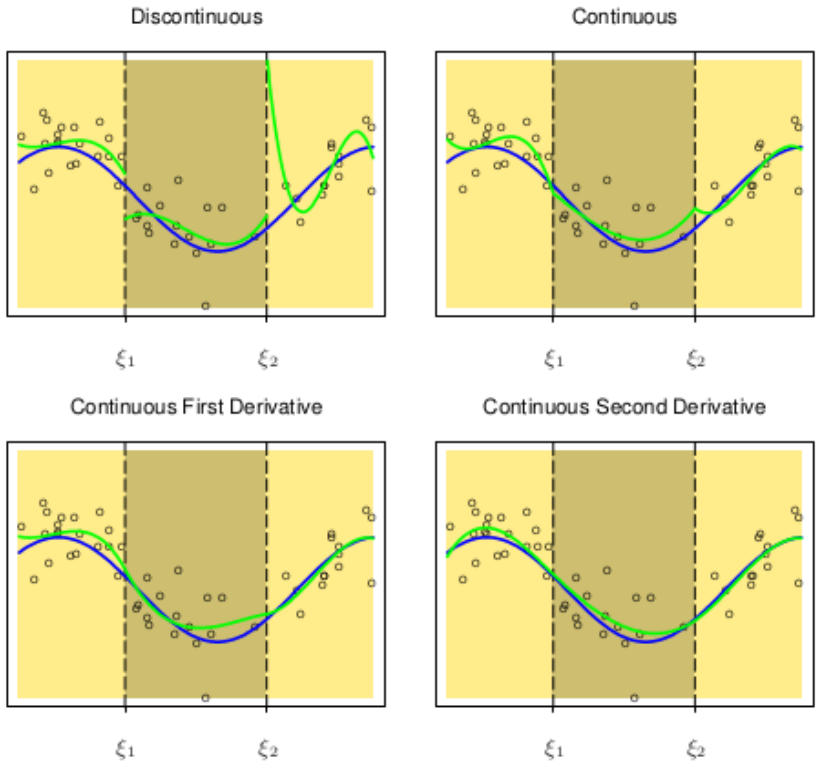
- Identify $K+1$ disjoint regions by specifying K knots ξ_k.
- Fit a polynomial in each region.

Continuity

- Often desirable to have continuity at the knots, $f\left(\xi_k^-\right) = f\left(\xi_k^+\right)$.
- This can be achieved by adding constraints on parameters and optimizing the loss accordingly.
- Alternatively, we can use basis functions that incorporate these constraints and thus use regular linear model fitting.

Cubic spline

- Increase smoothness by increasing order of the polynomial at the knots.
- This can be enforced by using appropriate basis functions.
- Continuous second derivatives is the most typically used order.



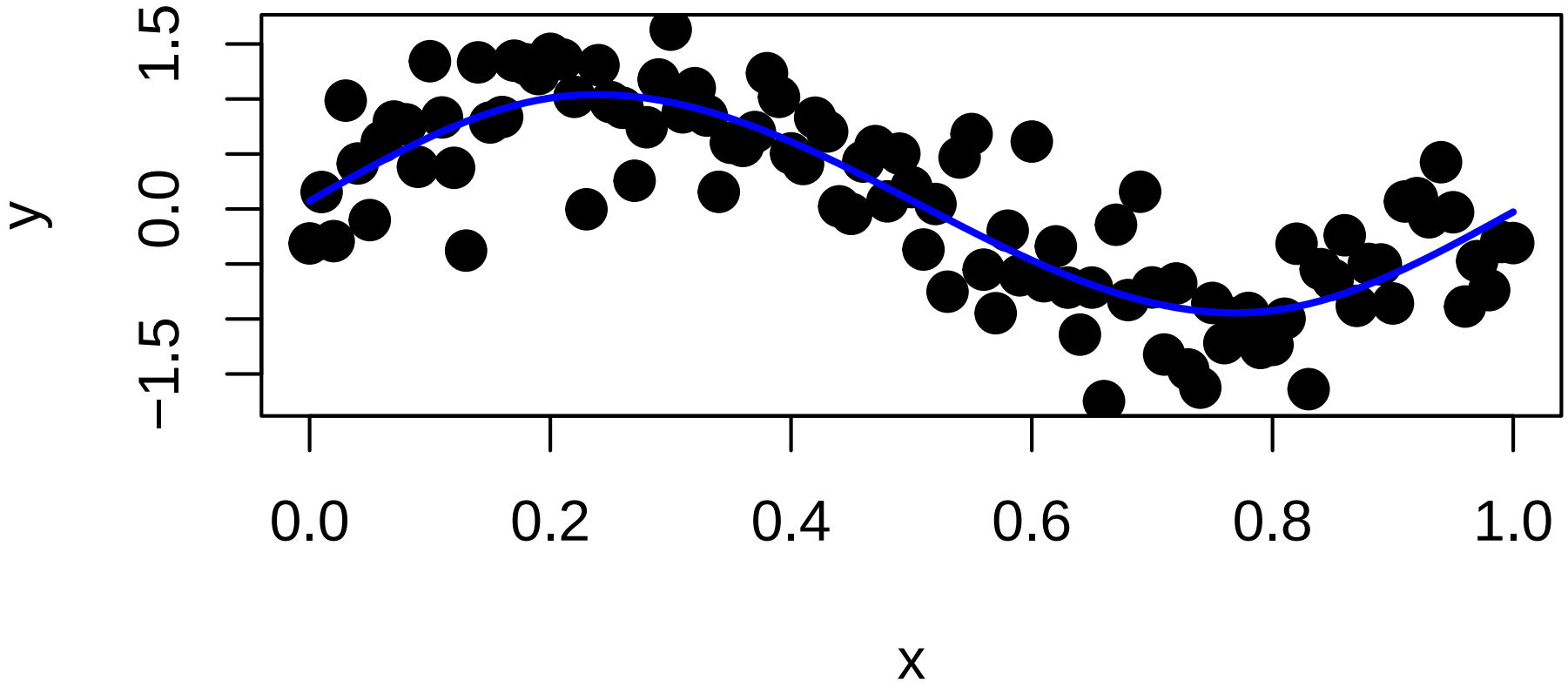
Plots show piecewise cubic polynomials; bottom right is a *cubic spline*.

Parameters

- While its possible to specify the knots, it is more common to let the data determine their positions (e.g., sample quantiles).
- Thus, two basic choices: degree of polynomial and number of knots (also knows as the degrees of freedom)
- The degree is typically cubic and we need to choose the number of degrees of freedom.

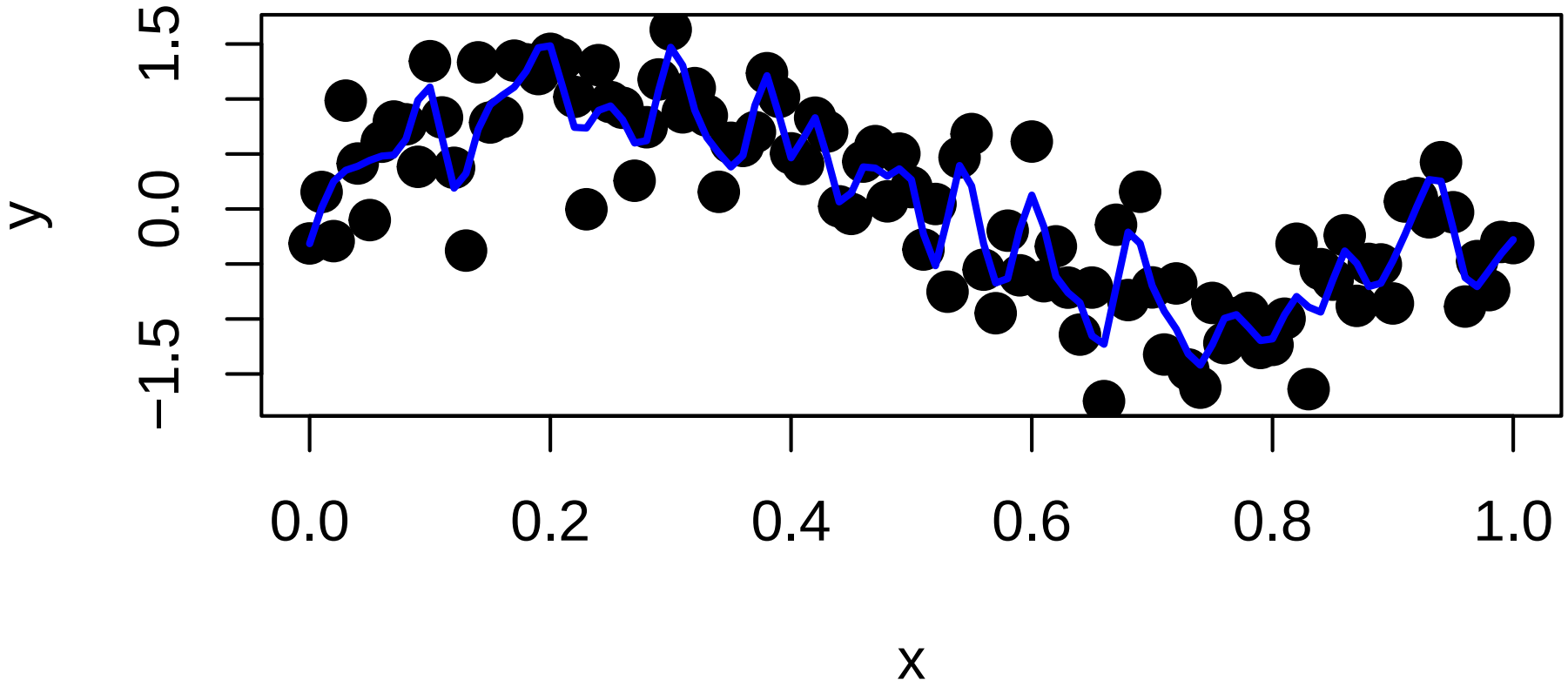
Cubic splines

With 5 knots.



Cubic splines

With 50 knots.



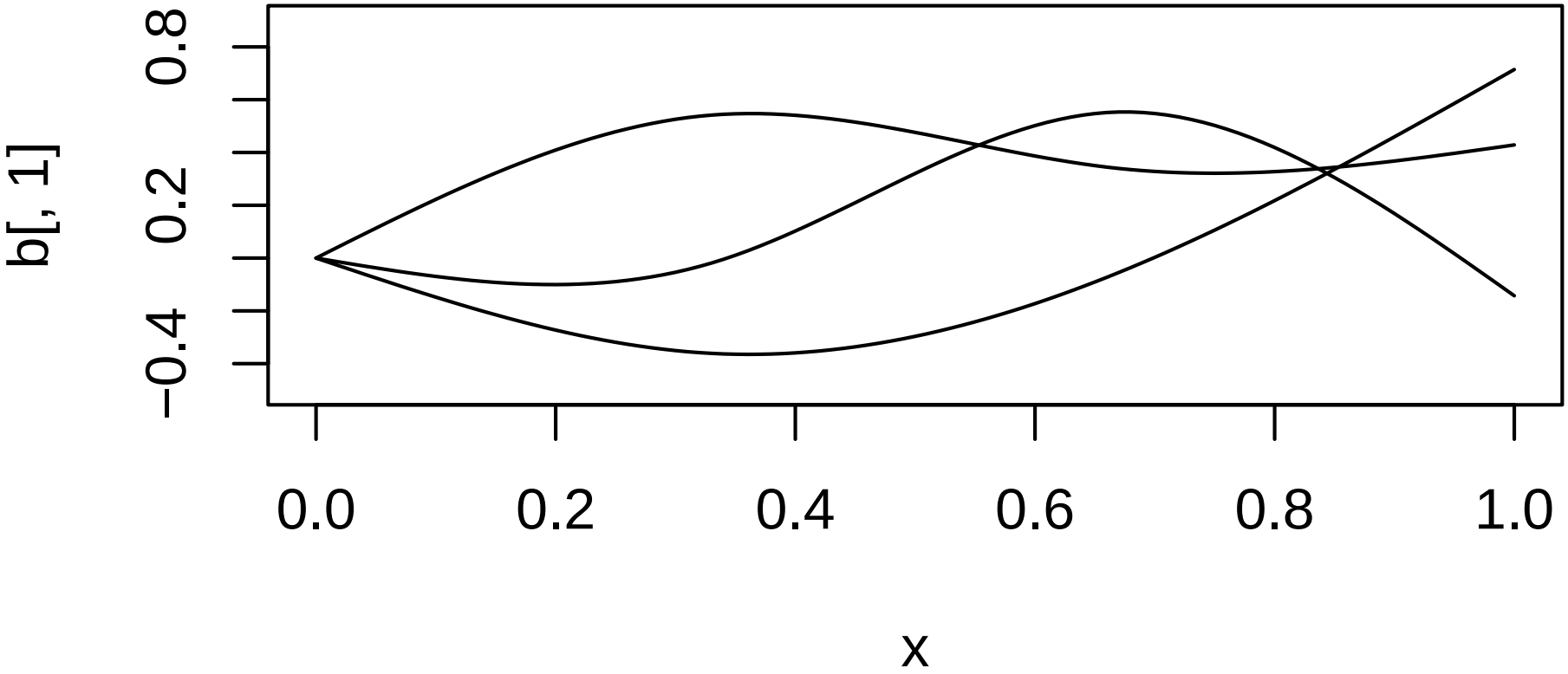
Natural cubic splines

- The polynomials fit beyond the boundary knots can behave wildly.
- Additional constraints: function is linear beyond the boundary knots.
- The constraint is enforced by the choice of appropriate basis functions.
- Given these additional constrains, a natural cubic spline with K knots has K basis functions.
- Since the constraint is enforced by the basis functions, we fit a standard linear model.

Natural cubic splines

With 3 knots.

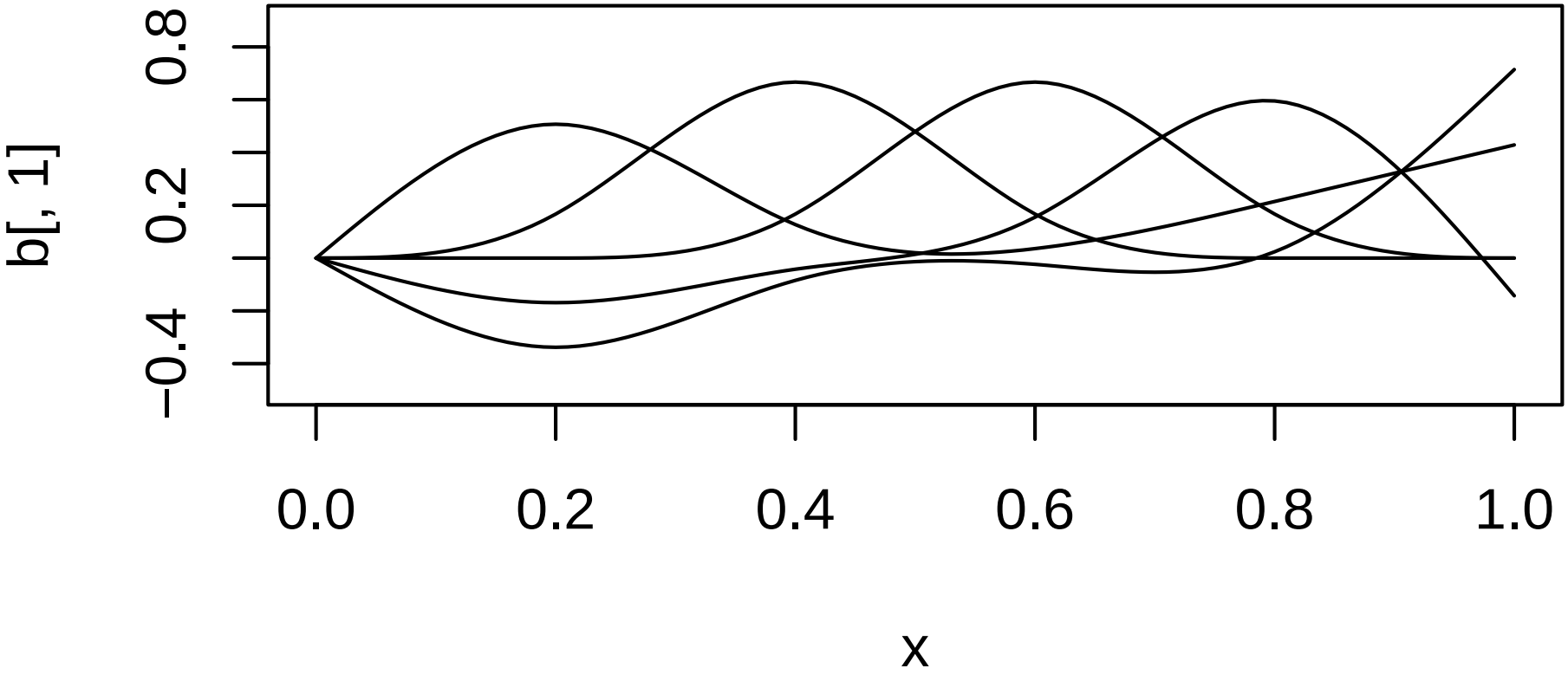
Basis functions



Natural cubic splines

With 5 knots.

Basis functions



Smoothing splines

- Instead of choosing the number of knots, use N, corresponding to training points.
- To limit complexity, shrink the coefficients when fitting the linear models.

Smoothing splines

Shrinking usually by penalizing the second derivative of the function,

$$\text{RSS}(f, \lambda) = \sum_{i=1}^{N} \{y_i - f(x_i)\}^2 + \lambda \int \{f''(t)\}^2 \, dt,$$

where λ is a smoothing parameter. We penalize the square of the second derivative because we want a smooth, rather than wiggly, function.

Remarkably, the unique minimizer is a natural cubic spline with knots at the unique values of the x.

Learning the GAM

Now consider learning a full GAM model,

$$g(\mathrm{E}(Y \mid x_1, x_2, \ldots, x_n)) = \alpha + f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n),$$

for features $X_1, \ldots, X_n$.

Using cubic splines for regression and optimizing the penalized residual sum of squares, our loss function is

$$\mathrm{PRSS}(\alpha, f_1, f_2, \ldots, f_n) = \sum_{i=1}^{N} \left(y_i - \alpha - \sum_{j=1}^{N} f_j(x_{ij}) \right)^2 + \sum_{j=1}^{n} \lambda_j \int f_j''(t_j)^2 \, dt_j,$$

where the $\lambda_j \geq 0$ are tuning parameters. Note that there is a λ_j per each feature.

Learning the GAM

- The minimizer of the loss function is an *additive cubic spline model*.
- Each f_j is a cubic smoothing spline in each of the variables.
- Without further constraints, the model is not unique – α is not identifiable: we can add/subtract any constants to f_j and adjust α accordingly.
- Question: Why is this not an issue with linear models?

Learning the GAM

- The minimizer of the loss function is an *additive cubic spline model*.
- Each f_j is a cubic smoothing spline in each of the variables.
- Without further constraints, the model is not unique – α is not identifiable: we can add/subtract any constants to f_j and adjust α accordingly.
- Question: Why is this not an issue with linear models? We cannot change β_j so that it adds a constant to $\beta_j X_j$.
- Standard convention: assume that $\sum_1^N f_j(x_{ij}) = 0 \forall j$ - the functions average zero over the data.
- Then $\hat{\alpha} = \text{ave}(y_i)$ for regression.

Backfitting

An interative algorithm for finding the minimizer.

Backfitting for regression

1. Initialize: $\hat{\alpha} = \frac{1}{N}\sum_1^N y_i, \quad \hat{f}_j \equiv 0, \forall j.$
2. Cycle: $j = 1, 2, \ldots, n, \ldots, 1, 2, \ldots, n, \ldots,$
   1. $\hat{f}_j \leftarrow \mathcal{S}_j \left[\left\{ y_i - \hat{\alpha} - \sum_{k \neq j} \hat{f}_k (x_{ik}) \right\}_1^N \right].$
   2. $\hat{f}_j \leftarrow \hat{f}_j - \frac{1}{N}\sum_{i=1}^N \hat{f}_j (x_{ij}).$

until the functions $\hat{f}_j$ change less than a prespecified threshold.

Step 1:

- Set $\hat{\alpha} = \text{ave}(y_i)$, and it never changes.
- Initialize f_j.

Backfitting

Backfitting for regression

1. Initialize: $\hat{\alpha} = \frac{1}{N}\sum_1^N y_i, \quad \hat{f}_j \equiv 0, \forall j.$
2. Cycle: $j = 1, 2, \ldots, n, \ldots, 1, 2, \ldots, n, \ldots,$
   1. $\hat{f}_j \leftarrow \mathcal{S}_j \left[\left\{ y_i - \hat{\alpha} - \sum_{k \neq j} \hat{f}_k (x_{ik}) \right\}_1^N \right].$
   2. $\hat{f}_j \leftarrow \hat{f}_j - \frac{1}{N}\sum_{i=1}^N \hat{f}_j (x_{ij}).$

until the functions $\hat{f}_j$ change less than a prespecified threshold.

Step 2.1:

- Apply a cubic smooth spline $\mathcal{S}_j$ to fit the current residuals, obtained using all features but X_j, from X_j.
- Use the corresponding penalty λ_j.
- The previous version of f_j is replaced.

Backfitting

Backfitting for regression

1. Initialize: $\hat{\alpha} = \frac{1}{N}\sum_1^N y_i, \quad \hat{f}_j \equiv 0, \forall j.$
2. Cycle: $j = 1, 2, \ldots, n, \ldots, 1, 2, \ldots, n, \ldots,$
   1. $\hat{f}_j \leftarrow \mathcal{S}_j \left[\left\{ y_i - \hat{\alpha} - \sum_{k \neq j} \hat{f}_k (x_{ik}) \right\}_1^N \right].$
   2. $\hat{f}_j \leftarrow \hat{f}_j - \frac{1}{N}\sum_{i=1}^N \hat{f}_j (x_{ij}).$

until the functions $\hat{f}_j$ change less than a prespecified threshold.

Step 2.2:

- Center f_j after fitting it.
- In theory, not needed as smoothing spline fit to a mean-zero response has mean zero;
- Due to machine rounding, however, useful to do it.

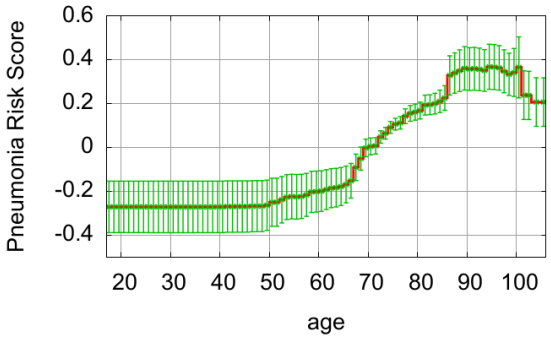
Backfitting

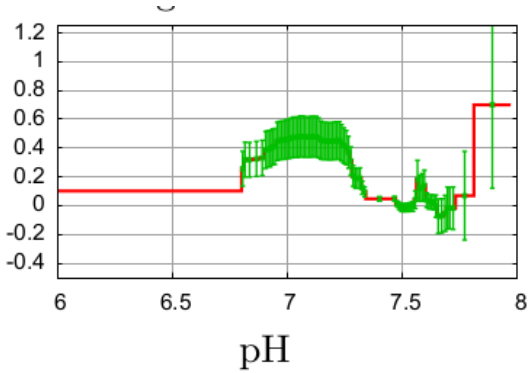
- Simple and modular: can choose a fitting method appropriate for each input variable.
- Smoothing operators other than the smoothing cubic splines can be used, such as
  - local polynomial regression and kernel methods;
  - tree-based predictors;
  - surface smoothers for second or higher-order interactions.
- Can be adapted for logistic regression and other GAMs.
- Multi-colinearity: sensitive to variable ordering.

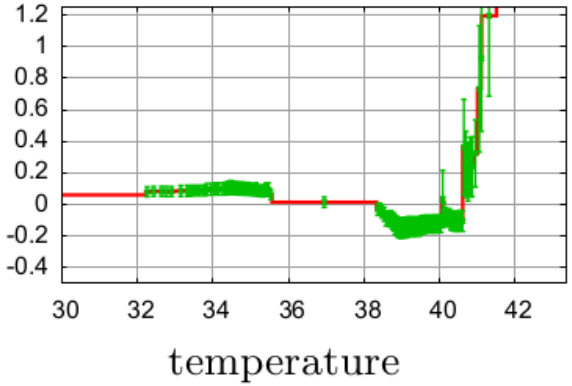
Table of Contents

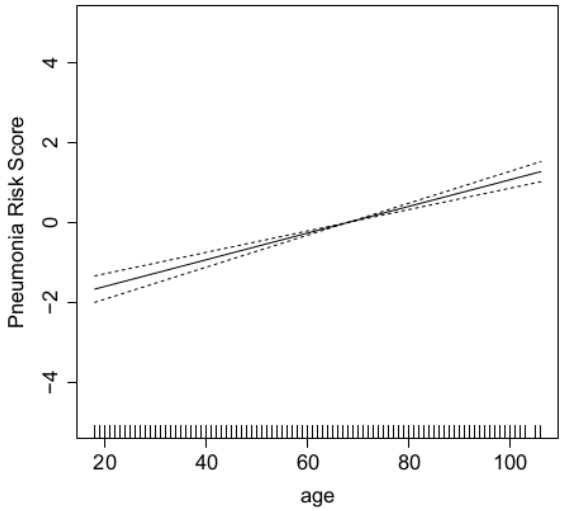
1. Case study: Pneumonia risk prediction
2. Learning with splines
3. Learning with trees
4. Summary

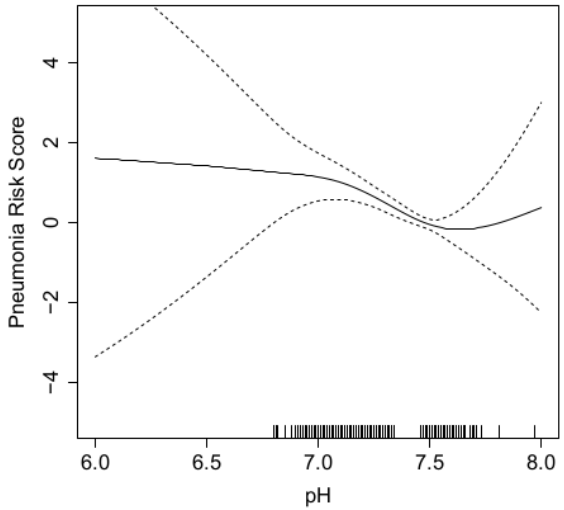
Splines versus tree-based

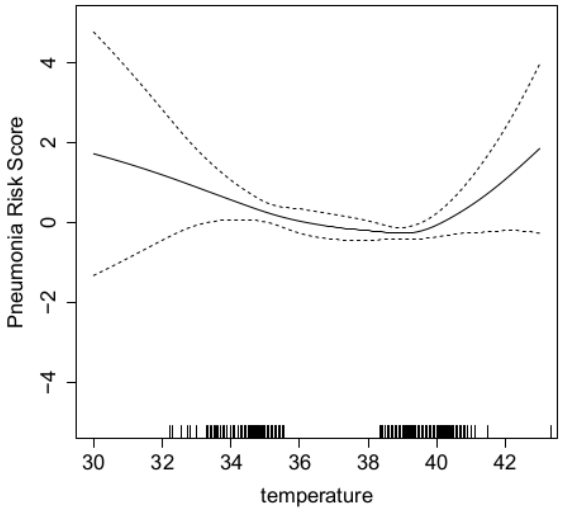












Splines versus tree-based

- The splines capture the basic trends: risk increases with age, pH risk is least around 7.6, and fever risk rises above 40 degrees, the splines miss detail learned by the trees.
- For example, the tree-based function for age is much more nuanced.
- The spline may not be properly modelling temperature in the normal range 36-38 degrees.
- The splines enforce smoothness and provide more principled error bars than tree-based methods.
- However, in a number of empirical studies, splines provided a poorer fit, resulting in higher RMSE or lower accuracy.

Splines versus tree-based

- Perhaps the smoothness constraint is inadequate in some cases.
- In order to fit discontinuities, a spline would need to use high-degree polynomials.

Not all phenomena are smooth

- Freezing and boiling occur at distinct temperatures,
- Human decision making processes: loans, admission to school, or administering medical procedures use discrete thresholds.

Explainable Boosting Machine

Video

- The Science Behind InterpretML: Explainable Boosting Machine
- https://youtu.be/MREiHgHgl0k?t=82

Learning

Gradient Boosting for Regression

$f_j \leftarrow 0$
for $m = 1$ to M do
  for $j = 1$ to n do
$\mathcal{R} \leftarrow \{x_{ij}, y_i - \sum_k f_k\}_1^N$
  Learn shaping function $S : x_j \rightarrow y$ using $\mathcal{R}$ as training dataset
$f_j \leftarrow f_j + \nu S$

- This is essentially identical to backfitting, with the difference being that, instead of replacing f_j with S we add νS to f_j.
- Importantly, this allows us to incorporate a learning rate ν.
- A low ν increases computation time but mitigates effects of variable ordering and so as to seek not to overshoot a good solution.

Learning

Gradient Boosting for Regression

- Adding S to f_j also allows us to use bagging at the different iterations, in order to reduce the extra variance due to trees.
- In backfitting, we re-build f_j for each feature from scratch in each round, so the shape function needs to be expressive to capture a complex function.
- In gradient boosting, many different S are combined to form f_j and thus they can be simple, such as decision tree stumps.

Interpreting an ensemble of trees?

- The complex ensembles of trees are not interpretable.
- They are thus, converted to lookup tables which can be visualized.
- This is done by partitioning the domain of X_j into regions and keeping a single value of the trees for that region.
- Thus, unlike spline, these functions can be discontinous as they are based on binning.
- Prediction is extremely fast as it involves additions and lookups inside of the feature functions.
- The lookup tables require little memory and are thus convenient for deployment in production.
- Essentially, what EBM does is to learn, in a supervised way, a discretization of features, and appropriate constants for the bins

ExplainableBoostingClassifier in interpretml package

Default values for some parameters

- `validation_size=0.15`
- `max_bins=1024`
- `learning_rate=0.01`
- `interactions=0.95`
- `max_rounds=25000`
- `early_stopping_rounds=50`

Enforce monotonicity

We might improve interpretability by enforcing monotonicity in a (main) term

Parameter: `monotone_constraints`

> *This parameter allows you to specify monotonic constraints for each feature's relationship with the target variable during model fitting. However, it is generally recommended to apply monotonic constraints post-fit using the monotonize function rather than setting them during the fitting process. This recommendation is based on the observation that, during fitting, the boosting algorithm may compensate for a monotone constraint on one feature by utilizing another correlated feature, potentially obscuring any monotonic violations.*

Enforce monotonicity

- Function: `monotonize(term, increasing='auto', passthrough=0.0)`
- Parameter `passthrough` – monotonization can change the mean response of the model. If passthrough is set to 1.0 then any change to the mean response made by monotonization will be passed through to `self.intercept_`. Values between 0 and 1 will result in that percentage being passed through.
- What should we expect regarding train/test after modifying a model, versus somehow preprocessing and retraining it?

Table of Contents

1 Case study: Pneumonia risk prediction

2 Learning with splines

3 Learning with trees

4 Summary

Summary

Splines, EBM and more

- Neural additive model (NAM) (Agarwal et al., 2021),
- GAMI-Net (Yang et al., 2021a), and
- Enhanced explainable neural network (ExNN) (Yang et al., 2021b
- ...

How to choose?

- Different GAMs can have similar accuracy but very different shape functions
- EBM and neural nets: slow training, especially with high n (fast prediction with EBMs, however).
- With sufficient data more flexible models may be better for exploration, identifying anomalies, such as imputation effects, biases etc.
- That is, they have higher "data fidelity"; and hence might generalize worse if data not representative.

Summary

- GAM/GA2M combine accuracy and interpretability.
- Of course, sometimes this is not enough. There might be, for example, interactions of degree three or higher.
- For instance, age might interact with cancer differently depending on a third feature.
- Can be used for initial data exploration, to look for anomalies, biases.

Summary

Data exploration

"Intelligibility Helps Create New Knowledge":
https://www.youtube.com/watch?v=MREiHgHgl0k&t=198s

> *GAMs are often used on raw medical records or other complex data types, and these datasets are likely to benefit from troubleshooting. Using a GAM, we might find counterintuitive patterns; e.g., . . . , asthma patients fared better than non-asthma patients in a health outcomes study. . . Medical records are notorious for missing important information or providing biased information such as billing codes. Could a GAM help us to identify important missing confounders, such as retirement effects or special treatment for asthma patients? Could GAMs help us reconcile medical records from multiple data storage environments?*

Interpretation

- Because GA^{2}M are interpretable, tempting to interpret them causally.
- While they can explain their predictions, they are based on association.
- If features were added to or subtracted and the model retrained, the graphs for some terms that had remained in the model would change because of correlation with the features added or subtracted.
- Although details of some of the shape plots are suggestive –e.g., jumps in risk near retirement age–, the model cannot answer whether they are due to:
  - overfitting;
  - correlation with other variables;
  - interaction with other variables;
  - correlation or interaction with missing variables;
  - due to true underlying phenomena such as retirement.

Interpretation

- Causal considerations requires assumptions and knowledge beyond the data and the model itself.
- In the pneumonia risk example, the model was edited manually.
- Precisely because the model is interpretable, users could understand what the model is doing and be safe against unwanted rules.
- Identifying and acting on these rules relied on external knowledge.